Online Resource 1

Gated Cluster-Pooled Sign Constraints for Multi-Output Sparse Regression

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This supplement supplies mathematical details, complete study protocols, extended results and computational checks for the main article. The previously verified results are retained, with a further variance-convention sensitivity comparison in Section S10, a Monte Carlo check of the pooling precision formula in Section S2 and the jointly tuned eQTL protocol in Section S9. The record of historical conventions and corrections is kept with Online Resource 2. Figures were produced with Python and Matplotlib.

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S1 Conditions and proof for the on-support gate result

The result below concerns the sign matrix produced by the date-score gate, not the support or distribution of the fitted coefficient matrix. It allows contemporaneous response dependence and random response clusters, with a structural assumption on which clusters can be selected asymptotically. The number of predictors and responses is fixed.

Let $(\boldsymbol{x}_t, \boldsymbol{y}_t)$ denote population-centered observations and let $\boldsymbol{\beta}_{\text{true}}$ be the coefficient matrix in $\boldsymbol{y}_t = \boldsymbol{\beta}_{\text{true}}\boldsymbol{x}_t + \boldsymbol{\varepsilon}_t$. For any nonempty response subset $\mathcal{C}$, write $m = |\mathcal{C}|$, $\bar{y}_{t,\mathcal{C}} = m^{-1}\sum_{k\in\mathcal{C}} y_{tk}$, and define

$$q_j = E[x_{tj}^2], \qquad b_{j,\mathcal{C}} = \frac{E[x_{tj}\bar{y}_{t,\mathcal{C}}]}{q_j}, \qquad \bar{r}_{t,j,\mathcal{C}} = \bar{y}_{t,\mathcal{C}} - b_{j,\mathcal{C}}x_{tj}. \tag{S1}$$

Here $b_{j,\mathcal{C}}$ is a population marginal slope, not an individual partial coefficient. Define the population score scale

$$\omega_{j,\mathcal{C}}^2 = q_j^{-2}E[x_{tj}^2\bar{r}_{t,j,\mathcal{C}}^2]. \tag{S2}$$

With fixed analysis scales this is the marginal slope's influence variance. When column scales are estimated from the same observations, additional scale-estimation terms can enter the slope's centered limiting distribution. The result below uses only the displayed positive score scale and makes no calibrated-inference claim for that sample-normalized slope. Let $\mathfrak{C}$ be a deterministic finite family of nonempty response subsets. A pair $(j,\mathcal{C})$ is called relevant if $\beta_{\text{true},kj} \neq 0$ for at least one $k \in \mathcal{C}$.

Conditions.

(A1) Sampling and exogeneity. The dates $(\boldsymbol{x}_t, \boldsymbol{\varepsilon}_t)$ are independent and identically distributed, $E[\boldsymbol{x}_t] = \mathbf{0}$, and $E[\boldsymbol{\varepsilon}_t \mid \boldsymbol{x}_t] = \mathbf{0}$. The matrix $\boldsymbol{\beta}_{\text{true}}$ and dimensions N, M do not vary with T. Response errors within a date may be dependent.

(A2) Moments, centering and analysis units. Every component of $\boldsymbol{x}_t$ and $\boldsymbol{\varepsilon}_t$ has a finite fourth moment, and $\boldsymbol{\Sigma}_F = E[\boldsymbol{x}_t\boldsymbol{x}_t^\top]$ is positive definite. Only complete observations are used. If a column is standardized, its population variance is positive, and $\boldsymbol{x}_t, \boldsymbol{y}_t$ and $\boldsymbol{\beta}_{\text{true}}$ denote the variables and coefficients in the corresponding population-standardized units. If a column is not standardized, it retains its specified fixed measurement units. Thus (A3), (A4) and (A6) refer to pooled marginal slopes in these limiting analysis units. Positive column rescaling preserves individual partial signs, but need not preserve a pooled marginal sign. The empirical columns are centered by their sample means. Write $\widehat{s}_{x,j}, \widehat{s}_{y,k}$ for their further empirical scale factors in the limiting units: each is one for an unstandardized column and its sample standard deviation for a standardized column. Hence $\widehat{s}_{x,j} = 1 + O_p(T^{-1/2})$ and $\widehat{s}_{y,k} = 1 + O_p(T^{-1/2})$, as follows from the fourth moments and positive population variances.

(A3) Admissible clusters and coherence. Let $\widehat{\mathcal{P}}_T$ be the supplied or estimated partition. The event $\mathcal{E}_T = \{\widehat{\mathcal{P}}_T \subseteq \mathfrak{C}\}$ has probability tending to one. Within every $\mathcal{C} \in \mathfrak{C}$ and for every predictor j, all nonzero $\beta_{\text{true},kj}$ have the same sign. For a fixed supplied partition, $\mathcal{E}_T$ can hold identically.

(A4) Nondegenerate marginal signals. For constants $\delta > 0$ and $0 < c < C < \infty$, every relevant pair with $\mathcal{C} \in \mathfrak{C}$ satisfies $|b_{j,\mathcal{C}}| \geq \delta$ and $c \leq \omega_{j,\mathcal{C}}^2 \leq C$. This is a population signal condition; it does not assume that an estimated gate statistic has already passed its threshold.

(A5) Gate and threshold. Use the complete-data, equal-weight statistic in Equation (4). The deterministic threshold satisfies $\tau_T \geq 0$ and $\tau_T = o(\sqrt{T})$; any fixed finite threshold is included. No penalty-rate assumption is imposed.

(A6) Marginal sign agreement. For every relevant $(j,\mathcal{C})$ with $\mathcal{C} \in \mathfrak{C}$ and every nonzero $\beta_{\text{true},kj}$ in that cluster, $\operatorname{sign}(b_{j,\mathcal{C}}) = \operatorname{sign}(\beta_{\text{true},kj})$.

Under exogeneity the marginal slope is

$$b_{j,\mathcal{C}} = \frac{1}{m\,\Sigma_{F,jj}} \sum_{k\in\mathcal{C}} \sum_{\ell=1}^{M} \beta_{\text{true},k\ell}\Sigma_{F,j\ell}. \tag{S3}$$

With uncorrelated predictors it is the average partial coefficient, so (A3) implies (A6). With correlated predictors, the other terms can reverse its sign. Condition (A3) is an assumption about the clustering procedure under the data-generating law, not a consequence of positive response correlations. Conditions (A3)–(A6) permit partially shared supports, but exclude cancellation or sign reversals in the relevant population pooled slopes.

Lemma 1 (a sufficient condition for marginal-sign agreement). Assume (A1), (A2) and the coherence part of (A3) for a relevant pair $(j,\mathcal{C})$, with common sign $s \in \{-1,+1\}$ on its nonzero $\beta_{\text{true},kj}$. If

$$\Sigma_{F,jj} \sum_{k\in\mathcal{C}} |\beta_{\text{true},kj}| \;>\; \sum_{k\in\mathcal{C}} \sum_{\ell\neq j} |\beta_{\text{true},k\ell}|\,|\Sigma_{F,j\ell}|, \tag{S4}$$

then $\operatorname{sign}(b_{j,\mathcal{C}}) = s$ and $|b_{j,\mathcal{C}}|$ is at least the difference of the two sides of Equation (S4) divided by $m\,\Sigma_{F,jj}$.

Proof. Equation (S3) separates into own-effect and leakage terms,

$$m\,\Sigma_{F,jj}\,b_{j,\mathcal{C}} = \Sigma_{F,jj} \sum_{k\in\mathcal{C}} \beta_{\text{true},kj} + \sum_{k\in\mathcal{C}} \sum_{\ell\neq j} \beta_{\text{true},k\ell}\Sigma_{F,j\ell}.$$

By coherence every nonzero $\beta_{\text{true},kj}$ in the cluster has sign s. The first term therefore equals

$$s\,\Sigma_{F,jj} \sum_{k\in\mathcal{C}} |\beta_{\text{true},kj}|.$$

The triangle inequality bounds the absolute value of the second term by the right-hand side of Equation (S4). Multiplying by s gives $s\,m\,\Sigma_{F,jj}\,b_{j,\mathcal{C}}$ at least the difference of the two sides, which is positive. □

With uncorrelated predictors the right-hand side is zero and the condition holds for every relevant coherent pair. With $\boldsymbol{\Sigma}_F$ scaled to unit diagonal it reads $\sum_k |\beta_{\text{true},kj}| > \sum_k \sum_{\ell\neq j} |\beta_{\text{true},k\ell}|\,|\rho_{j\ell}|$, where $\rho_{j\ell}$ is the predictor correlation. The condition is sufficient and not necessary, because leakage terms of opposite sign can cancel. It is a cluster-summed, coefficient-weighted bound on the cross-covariances of one predictor, and so differs from the irrepresentable condition of Zhao and Yu (2006), which bounds a regression of the inactive design on the active design.

Proposition 1 (correct gate signs on the support). Let $\widehat{\boldsymbol{S}}_T$ be the date-score sign matrix with no practitioner overrides or unrestricted cells, and let

$$\mathcal{H}_T = \{\widehat{S}_{T,kj} = \operatorname{sign}(\beta_{\text{true},kj}) \text{ for all } (k,j) \text{ with } \beta_{\text{true},kj} \neq 0\}. \tag{S5}$$

Under (A1)–(A6), $\Pr(\mathcal{H}_T) \to 1$. This statement does not assert $\operatorname{supp}(\widehat{\boldsymbol{\beta}}_T) = \operatorname{supp}(\boldsymbol{\beta}_{\text{true}})$.

Proof. For a fixed pair $(j,\mathcal{C})$, define the empirical analysis columns

$$\widetilde{x}_{tj} = \frac{x_{tj} - \bar{x}_j}{\widehat{s}_{x,j}}, \qquad \widetilde{y}_{t,\mathcal{C}} = \frac{1}{m} \sum_{k\in\mathcal{C}} \frac{y_{tk} - \bar{y}_k}{\widehat{s}_{y,k}}.$$

The pooled slope is

$$\widehat{b}_{j,\mathcal{C}} = \frac{T^{-1}\sum_t \widetilde{x}_{tj}\widetilde{y}_{t,\mathcal{C}}}{T^{-1}\sum_t \widetilde{x}_{tj}^2}.$$

Finite fourth moments give finite variances for the products entering sample cross moments and sample variances. Their averages have $O_p(T^{-1/2})$ deviations from their expectations. Sample-mean corrections are $O_p(T^{-1})$. For a standardized column the population variance is one in the limiting units; taking the square root of its sample variance and then its reciprocal preserves the $O_p(T^{-1/2})$ rate. There are only finitely many columns. The numerator and denominator above therefore differ from $E[x_{tj}\bar{y}_{t,\mathcal{C}}]$ and $q_j > 0$ by $O_p(T^{-1/2})$, yielding $\widehat{b}_{j,\mathcal{C}} - b_{j,\mathcal{C}} = O_p(T^{-1/2})$.

Write $\widehat{r}_{t,j,\mathcal{C}} = \widetilde{y}_{t,\mathcal{C}} - \widehat{b}_{j,\mathcal{C}}\widetilde{x}_{tj}$ and $\widehat{q}_j = T^{-1}\sum_t \widetilde{x}_{tj}^2$. With complete observations, the score and denominator in Equation (4) are $m\widetilde{x}_{tj}\widehat{r}_{t,j,\mathcal{C}}$ and $mT\widehat{q}_j$. Therefore its screening variance statistic satisfies

$$T\widehat{V}_{j,\mathcal{C}} = \frac{T}{T-1}\,\frac{T^{-1}\sum_t \widetilde{x}_{tj}^2\widehat{r}_{t,j,\mathcal{C}}^2}{\widehat{q}_j^2} \;\longrightarrow_p\; \omega_{j,\mathcal{C}}^2. \tag{S6}$$

For completeness, expand the squared fitted residual around $\bar{r}_{t,j,\mathcal{C}}$. Every correction contains a sample mean, $\widehat{b} - b$, or $\widehat{s}^{-1} - 1$ tending to zero, multiplied by an empirical polynomial moment of total degree at most four that is $O_p(1)$ under (A2). The remaining average converges to $E[x_{tj}^2\bar{r}_{t,j,\mathcal{C}}^2]$ by the law of large numbers. This establishes the positive finite limit of $T\widehat{V}$ needed for gate consistency. When column normalization is estimated, it does not identify $\widehat{V}$ as the full variance of the sample-normalized slope.

There are at most $M(2^N - 1)$ candidate pairs. Since this number is fixed, the slope and score-scale convergences hold simultaneously over $\mathfrak{C}$. For every relevant pair, (A4) then gives

$$\frac{|t_{j,\mathcal{C}}|}{\sqrt{T}} = \frac{|\widehat{b}_{j,\mathcal{C}}|}{\sqrt{T\widehat{V}_{j,\mathcal{C}}}} \longrightarrow_p \frac{|b_{j,\mathcal{C}}|}{\omega_{j,\mathcal{C}}} > 0, \qquad \operatorname{sign}(\widehat{b}_{j,\mathcal{C}}) = \operatorname{sign}(b_{j,\mathcal{C}}) \quad \text{with probability tending to one.}$$

Because $\tau_T/\sqrt{T} \to 0$, all relevant pairs pass the gate simultaneously with probability tending to one. On $\mathcal{E}_T$, every selected cluster is one of the pairs already controlled; (A6) identifies its direction with every nonzero partial coefficient it contains. A union bound and $\Pr(\mathcal{E}_T^c) \to 0$ establish $\Pr(\mathcal{H}_T) \to 1$. No conditioning argument that treats an estimated cluster as fixed is used. □

Corollary 1 (a population Ward condition). Suppose (A1)–(A2) hold and every response has positive population variance. Let $\boldsymbol{R}$ be the population response correlation matrix and $\widehat{\boldsymbol{R}}_T$ its complete-data sample counterpart. For disjoint, nonempty response sets A, B, define $\boldsymbol{e}_A = |A|^{-1}\mathbf{1}_A$ and the squared Ward merge height

$$H^2(A, B; \boldsymbol{R}) = \frac{2|A||B|}{|A| + |B|}(\boldsymbol{e}_A - \boldsymbol{e}_B)^\top \boldsymbol{R}(\boldsymbol{e}_A - \boldsymbol{e}_B). \tag{S7}$$

Here $\mathbf{1}_A$ is the N-vector indicating membership in A. This is twice the increase in within-cluster sum of squares for unit-length response vectors with Gram matrix $\boldsymbol{R}$; for two singletons it equals $2(1 - R_{kl})$. Thus it uses the chord-distance Ward convention of Section 2.5.

Fix $\kappa \in (0, 1)$. Consider every population Ward merge sequence that starts at singletons and, at each step, chooses any pair attaining the smallest current value of H^2. Denote this finite set of sequences by $\mathcal{L}(\boldsymbol{R})$ and their merge heights by $h_{\ell,s}$, $s = 1, \ldots, N-1$. Assume that:

1. every sequence has $h_{\ell,N-1} > 0$ and cutting at $\kappa h_{\ell,N-1}$ gives the same partition $\mathcal{P}_\star$;
2. the cut has a positive margin from all merge heights,

$$\min_{\ell \in \mathcal{L}(\boldsymbol{R})} \min_{1 \le s < N} |h_{\ell,s} - \kappa h_{\ell,N-1}| > 0.$$

Then the sample chord-distance Ward partition cut at κ times its own final merge height satisfies

$$\Pr(\widehat{\mathcal{P}}_T = \mathcal{P}_\star) \longrightarrow 1. \tag{S8}$$

If each block of $\mathcal{P}_\star$ has coherent nonzero partial signs, condition (A3) follows with $\mathfrak{C} = \mathcal{P}_\star$. If (A4)–(A6) also hold for those blocks, Proposition 1 therefore applies to this sample-estimated Ward partition.

Proof. The law of large numbers, positive response variances and fixed N give $\widehat{\boldsymbol{R}}_T \to_p \boldsymbol{R}$ entrywise. For each pair A, B, $H^2(A, B; \boldsymbol{R})$ is a continuous function of $\boldsymbol{R}$. Only finitely many partitions and cluster pairs can occur. At any state reachable by a population minimum-cost sequence, the gap between a minimum-cost pair and any strictly more costly pair is positive. Taking the minimum of the finitely many positive gaps shows that, for matrices sufficiently close to $\boldsymbol{R}$, a sample minimizer at any such state must be one of its population minimizers. States without a strictly more costly pair require no gap. Induction over the merges shows that every sample sequence then belongs to $\mathcal{L}(\boldsymbol{R})$, although its choices among population ties may vary with the sample.

Along any fixed sequence, each height converges to its population value. This convergence is uniform over the finite set $\mathcal{L}(\boldsymbol{R})$. The positive cut margin preserves, for every merge, whether its height is below or above the fraction of the final height. Each possible sequence therefore gives the same sample cut partition $\mathcal{P}_\star$ once $\widehat{\boldsymbol{R}}_T$ is sufficiently close to $\boldsymbol{R}$. This proves Equation (S8). Coherence of the blocks gives the remaining part of (A3), and Proposition 1 gives the last claim. □

The condition permits ties only when their alternative trees agree at the chosen cut. A unique minimum at every population merge gives a unique population tree; a positive final height and the cut margin are still needed. Neither condition is inferred from observed positive correlations. The result concerns fixed dimensions and a fixed cut fraction, and does not apply to the archived squared-distance heuristic or to arbitrary pairwise-missing correlations.

Example (coherent Gaussian blocks). Let $G \geq 2$ blocks each have size $m \geq 2$, set $N = Gm$ and $M = G$, and take independent dates with

$$\boldsymbol{x}_t \sim N(\boldsymbol{0}, \boldsymbol{I}_G), \qquad \boldsymbol{\varepsilon}_t \sim N(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_N), \qquad \boldsymbol{x}_t \perp \boldsymbol{\varepsilon}_t,$$

where $a, \sigma > 0$. For $k \in \mathcal{C}_g$, let

$$y_{tk} = a\, s_g x_{tg} + \varepsilon_{tk}, \qquad s_g \in \{-1, +1\}. \tag{S9}$$

The signs may differ between blocks. Within a block the population response correlation is $\rho = a^2/(a^2 + \sigma^2) \in (0, 1)$; between blocks it is zero.

For two disjoint subsets of sizes u, v from the same block, Equation (S7) gives $H^2 = 2(1 - \rho)$. For subsets from different blocks it gives

$$H^2 = 2(1 - \rho) + \frac{4\rho uv}{u + v} > 2(1 - \rho).$$

Consequently all within-block merges occur before any between-block merge, regardless of ties. Once the blocks have formed, their centroids are mutually orthogonal in the Gram representation and have squared norm $\rho + (1-\rho)/m$. For unions of respectively r and s whole blocks, Equation (S7) then gives $H^2 = 2\{1 + (m-1)\rho\}$, independently of r, s. Thus every population sequence has within-block heights $h_W = \sqrt{2(1 - \rho)}$ and between-block heights $h_B = \sqrt{2\{1 + (m-1)\rho\}}$, including its final height. Every fixed cut fraction satisfying

$$\sqrt{\frac{1 - \rho}{1 + (m-1)\rho}} < \kappa < 1 \tag{S10}$$

has a positive margin and yields the true block partition under every tie choice. Corollary 1 therefore establishes (A3) for this data-generating model.

For each relevant pair $(g, \mathcal{C}_g)$, the population pooled slope is $b_{g,\mathcal{C}_g} = a s_g$, $E[x_{tg}^2] = 1$, and its influence variance is $\omega^2_{g,\mathcal{C}_g} = \sigma^2/m$. Conditions (A4) and (A6) follow, and all Gaussian moment

and exogeneity conditions hold. With any threshold sequence satisfying (A5), the learned Ward gate has correct signs on the nonzero coefficients with probability tending to one. This example establishes a nonempty model class; it is not a claim that the simulation designs or yeast data satisfy these separation and sign conditions, nor a support or prediction guarantee for the fitted estimator.

All responses in this example have the same variance $a^2 + \sigma^2$. Population standardization therefore leaves the response correlation geometry and partial signs unchanged, dividing the relevant pooled slopes by $\sqrt{a^2 + \sigma^2}$ and their influence variances by $a^2 + \sigma^2$. Training-sample standard deviations converge to these positive population scales. Continuity of the finitely many sample moments therefore preserves the pooled-slope limits and the positive limits of $T\widehat{V}_j$ needed for the on-support gate argument. This statement concerns eventual gate signs; it does not identify a centered limiting distribution after estimated scale normalization. In particular, standardization does not create a population marginal-sign reversal in this example.

Scope and limitations. The proof allows clusters estimated from the same observations when the event in (A3) has probability tending to one. Corollary 1 establishes that event under population partition and cut-margin conditions, with an explicit model in which the required sign agreement also holds. It does not establish those conditions for arbitrary response models, for the empirical application, or for dimensions increasing with sample size. Observed marginal-sign coherence cannot verify partial-sign coherence or (A6), since the partial-coefficient truth is unobserved.

For a prespecified pair computed in fixed measurement units or with fixed external scaling, a zero marginal slope and a nondegenerate score variance give an asymptotically standard normal date-score statistic by a separate central-limit argument. A fixed threshold then retains a nonvanishing fraction of such pairs. This fixed-scale statement is not asserted after jointly estimating heterogeneous response standard deviations. This observation is not a multiplicity result and does not transfer to adaptively selected clusters without further analysis. A zero partial coefficient can also have a nonzero pooled marginal slope, so a growing threshold alone need not identify the coefficient support.

The pooled pilot in Equation (S12) is not a cellwise consistent partial-coefficient pilot: with independent predictors and two response vectors $(1, 0)$ and $(0, 1)$ in one cluster, both pooled pilots converge to $1/2$. Their weights remain finite on the two zero cells, even before applying the fixed stabilizer. Hence the adaptive-penalty selection arguments of Zou (2006); Wang and Leng (2008) cannot be imported for this estimator. There is also no centered oracle claim for its penalized active coefficients: in a one-predictor positive-coefficient fit with zero target and $\alpha = 1$,

$$\widehat{b} = \widehat{b}_{\mathrm{OLS}} - \lambda_T W_T/(2\widehat{q}),$$

once the half-space is inactive. If $W_T \to w > 0$, a root-T centered OLS-equivalent limit would require this shrinkage bias to vanish at that scale. A post-selection refit is a different estimator and would still need its own selection justification.

Finally, fixed-span recency weighting has bounded effective temporal sample size, missing observations change the pooled estimating equation, and serial dependence changes its long-run score variance. None is covered by the iid, complete-data proposition. The corresponding finite-sample validation is reported separately.

S2 Monte Carlo check of the pooling precision formula

Equation (13) gives the conditional variance of a pooled slope under equal member slopes, equal marginal residual variances and a common marginal-residual correlation ρ_e. The check holds one predictor fixed and verifies three quantities by simulation: the ratio of the member-slope standard error to the pooled-slope standard error against $\sqrt{m/\{1 + (m-1)\rho_e\}}$, the mean of the date-score

variance of Equation (4) relative to the true conditional variance, and the mean of the archived independent-cell variance relative to the same truth.

Protocol. One predictor $x_t \sim N(0,1)$, $t = 1, \ldots, 80$, is drawn once (seed 20261025), centered and held fixed. For each cell (m, ρ_e) with $m \in \{1, 2, 4, 8, 16\}$ and $\rho_e \in \{0, 0.3, 0.6, 0.9\}$, the m responses are $y_{tk} = 0.2\, x_t + e_{tk}$ with $\boldsymbol{e}_t \sim N(\mathbf{0}, (1-\rho_e)\boldsymbol{I}_m + \rho_e \mathbf{1}\mathbf{1}^\top)$ independent over dates, in 20,000 replications. Each replication computes the pooled slope of Equation (2), the date-score variance, the independent-cell variance $\mathrm{SSR}/\{(Tm-1)D_j\}$, the m member slopes, and the date-score statistics of the pooled slope and of the first member slope. The precision ratio is the mean across members of the Monte Carlo standard deviation of a member slope divided by the Monte Carlo standard deviation of the pooled slope. The variance ratios divide the mean of each estimator by the Monte Carlo variance of the pooled slope. The retention columns are the proportions of replications in which the pooled and the singleton statistic exceed the threshold 2 in absolute value. The producer is `pooling_precision_check.py` and the saved table is `pooling_precision_check.csv`.

Table S1: Monte Carlo check of the pooling precision formula. Ratio is the empirical member-to-pooled standard-error ratio and Formula is $\sqrt{m/\{1+(m-1)\rho_e\}}$. Date and Indep. are the mean date-score and independent-cell variances divided by the Monte Carlo variance of the pooled slope. Pooled and Single are retention proportions at threshold 2 for the pooled and singleton gates with member slope 0.2, $T = 80$ and 20,000 replications per cell.

m	ρ_e	Ratio	Formula	Date	Indep.	Pooled	Single
1	0.0	1.000	1.000	0.959	0.983	0.440	0.440
2	0.0	1.424	1.414	0.982	1.009	0.720	0.445
4	0.0	1.998	2.000	0.972	0.996	0.946	0.443
8	0.0	2.831	2.828	0.971	0.996	0.999	0.439
16	0.0	3.993	4.000	0.976	0.999	1.000	0.439
1	0.3	1.000	1.000	0.980	1.006	0.444	0.444
2	0.3	1.241	1.240	0.980	0.772	0.607	0.440
4	0.3	1.455	1.451	0.984	0.528	0.737	0.442
8	0.3	1.597	1.606	0.964	0.317	0.815	0.438
16	0.3	1.702	1.706	0.976	0.181	0.860	0.440
1	0.6	1.000	1.000	0.976	1.000	0.448	0.448
2	0.6	1.119	1.118	0.968	0.618	0.520	0.435
4	0.6	1.195	1.195	0.972	0.354	0.580	0.449
8	0.6	1.241	1.240	0.981	0.192	0.609	0.446
16	0.6	1.262	1.265	0.964	0.098	0.617	0.441
1	0.9	1.000	1.000	0.972	0.996	0.445	0.445
2	0.9	1.026	1.026	0.976	0.524	0.466	0.449
4	0.9	1.039	1.040	0.970	0.266	0.477	0.446
8	0.9	1.048	1.047	0.995	0.138	0.471	0.440
16	0.9	1.050	1.050	0.972	0.068	0.480	0.444

Reading. The empirical ratio follows the formula at every cell, with a largest absolute deviation of 0.010 at $m = 2$, $\rho_e = 0$. The date-score variance averages between 0.959 and 0.995 of the true conditional variance. The shortfall is the usual finite-sample downward bias of a sandwich estimator at $T = 80$, which the $T/(T-1)$ multiplier does not remove, and it does not affect the positive score-scale limit used by Proposition 1. The independent-cell variance averages $1/\{1+(m-1)\rho_e\}$ of the truth, as Equation (13) predicts, so its gate statistic is inflated by $\sqrt{1+(m-1)\rho_e}$ whenever marginal residuals are correlated. The retention columns show the practical consequence for a weak signal: at $\rho_e = 0$ the pooled gate at threshold 2 retains the active slope in 0.95 of replications for $m = 4$ against 0.44 for the singleton gate, and the gain shrinks to 0.58 at $\rho_e = 0.6$ and to

0.48 at $\rho_e = 0.9$. The check uses a prespecified cluster with equal member slopes and makes no statement about learned clusters or unequal effects.

S3 Full benchmark protocol and baseline coefficient results

We now evaluate fitted coefficients and independent test predictions. Each of six designs has 20 replications (seeds 0–19), 80 training observations, 1,000 independent test observations, 12 predictors and 12 responses. The baseline has three true clusters of size four, three active predictors per cluster, shared supports and signs, independent Gaussian predictors and errors, and population $R^2 = 0.30$. Magnitudes are independent uniform draws on $[0.5, 1.5]$. The weak-signal case changes R^2 to 0.10. Partial support replaces one active predictor for alternate members of each cluster; mixed signs reverses one active sign for alternate members. The correlated case sets predictor equicorrelation and within-cluster error correlation to 0.6. The unequal case uses cluster sizes $(2, 4, 6)$ and multiplies baseline error standard deviations by a sequence from 0.5 to 2, so response signal shares then differ. Seeds are paired across methods and cases; Monte Carlo replication units are seeds, not response cells.

The matched HCGL variants share the same response-row penalty, $\alpha = 0.5$, $W = 1$ and zero target. They differ only in their sign rule: no constraints, singleton screening with exclusion, estimated-cluster screening with exclusion, or estimated-cluster screening with failed gates released. An additional known-cluster diagnostic supplies the true partition to both the gate and group weights; it is not a feasible competitor or an isolated change to the gate. FCGL has the cluster-by-predictor norm and no sign constraints, also with $\alpha = 0.5$ and $W = 1$. Cooperative Lasso uses the positive/negative group norms on the same estimated partition, with no additional elementwise penalty. Ordinary Lasso is unconstrained and fits responses separately, but selects one common penalty fraction by the response-averaged validation loss; it does not tune each response separately. UniLasso uses the full leave-one-out univariate features and nonnegative second stage of Chatterjee et al. (2025), including its intercept.

Three-fold validation within the training sample repeats all centering, scaling, Ward clustering and sign derivation. The cluster cut is fixed at 0.5 of the final Ward merge height. For signed methods the initial search selects $\tau \in \{0.75, 1.5, 2.5\}$ jointly with the penalty fraction to minimize squared prediction error in training-standardized response units. The initial common positive fraction grid is $\{1, 0.3, 0.1, 0.03, 0.01, 0.003, 0.001\}$. Each method uses its training-only zero-solution penalty scale; for HCGL/FCGL this is a conservative sufficient bound from the elementwise penalty, not an exact KKT boundary. An initial diagnostic extended UniLasso to $10^{-4}, 10^{-5}, 10^{-6}$ and zero. A subsequent boundary audit extends every method's penalty grid by 3×10^{-4}, 10^{-4}, 3×10^{-5}, 10^{-5}, 3×10^{-6}, 10^{-6} and zero; the signed threshold grid also includes 0, 3, 6 and ∞. All reported fitted results use this final grid on the same seeds and inner folds. These extensions followed the boundary diagnostics and were not part of the initial protocol. Zero penalty is constrained least squares, or the nonnegative second-stage fit for UniLasso. The infinite threshold means complete gate failure: exclusion produces the zero-slope model, while release leaves every coefficient unrestricted. No test outcome or coefficient truth selects a penalty, threshold or fit. Reported paired intervals describe Monte Carlo variation without adjusting for the post-inspection grid extensions.

Table S2 compares independent-test prediction errors.

Estimated-cluster exclusion has no general predictive advantage over the matched unconstrained HCGL. Its paired test-MSE differences (mean, Monte Carlo standard error) are -0.00252 (0.00190) in the baseline, 0.00055 (0.00041) under weak signal, -0.00159 (0.00502) with partial supports, 0.00080 (0.00078) with mixed signs, 0.01568 (0.00641) in the correlated case, and 0.00082 (0.00077) in the unequal case. The correlated case shows a material loss from exclusion; its paired t_{19} interval is $[0.00226, 0.02909]$. The wider grid reduces the earlier loss but does not eliminate it. Releasing failed gates gives MSE 0.795, versus 0.797 without constraints and 0.812 with exclusion. In that case, cooperative Lasso has MSE 0.781 and ordinary Lasso 0.786.

The cut rule often discovers nearly singleton groups: the mean number of clusters is 10.15 in

Table S2: Independent test MSE in training-standardized response units. Means and Monte Carlo standard errors (parentheses) over 20 paired replications per case. Lower is better. Partial denotes partial supports, Mixed mixed signs, Corr. correlated predictors and errors, and Unequal unequal cluster sizes and error variances. The last row uses known groups.

Method	Base	Weak	Partial	Mixed	Corr.	Unequal
Lasso	0.808 (0.011)	1.006 (0.011)	0.824 (0.012)	0.817 (0.014)	0.786 (0.016)	0.849 (0.015)
HCGL, unconstrained	0.805 (0.010)	1.007 (0.011)	0.818 (0.012)	0.807 (0.011)	0.797 (0.017)	0.834 (0.014)
HCGL, singleton/exclude	0.804 (0.010)	1.007 (0.011)	0.816 (0.012)	0.808 (0.012)	0.806 (0.017)	0.836 (0.014)
HCGL, pooled/exclude	0.802 (0.011)	1.007 (0.011)	0.817 (0.012)	0.808 (0.012)	0.812 (0.017)	0.835 (0.014)
HCGL, pooled/release	0.805 (0.010)	1.007 (0.011)	0.818 (0.012)	0.807 (0.011)	0.795 (0.017)	0.834 (0.014)
FCGL, unconstrained	0.800 (0.010)	1.010 (0.011)	0.811 (0.011)	0.805 (0.011)	0.792 (0.016)	0.831 (0.014)
Cooperative Lasso	0.807 (0.011)	1.006 (0.011)	0.824 (0.012)	0.817 (0.014)	0.781 (0.016)	0.848 (0.015)
UniLasso	0.807 (0.010)	1.020 (0.012)	0.822 (0.012)	0.813 (0.012)	0.824 (0.017)	0.837 (0.014)
HCGL, known/exclude	0.777 (0.010)	0.994 (0.011)	0.822 (0.011)	0.852 (0.011)	0.812 (0.017)	0.810 (0.013)

the baseline, 12 under weak signal, 11.90 with partial supports, 11.95 with mixed signs and 11.40 in the unequal case; it is 3 in the correlated case. Thus the small estimated-pooling effects do not demonstrate recovery of the latent groups. Known groups improve baseline prediction ($R^2 = 0.250$ versus 0.223 for unconstrained HCGL), but under mixed signs they reduce it to 0.183 versus 0.226. A known partition is insufficient when its coefficient directions disagree.
Table S3 gives the corresponding numerical results.

Table S3: Baseline fitted-coefficient results, mean (Monte Carlo standard error) over 20 seeds. Coefficient MSE uses all standardized coefficients without thresholding. Precision is the proportion of selected cells truly active; recovery uses true active cells; null selection uses true zero cells. The displayed selection tolerance is 10^{-4}, following the independent solver audit in Section S12.2.

Method	Coefficient MSE	Precision	Sign recovery	Null selection
Lasso	0.0065 (0.0003)	0.430 (0.029)	0.956 (0.011)	0.471 (0.035)
HCGL, unconstrained	0.0062 (0.0002)	0.391 (0.014)	0.968 (0.007)	0.519 (0.021)
HCGL, singleton/exclude	0.0061 (0.0002)	0.519 (0.025)	0.931 (0.013)	0.319 (0.031)
HCGL, pooled/exclude	0.0059 (0.0002)	0.557 (0.026)	0.929 (0.011)	0.275 (0.030)
HCGL, pooled/release	0.0062 (0.0002)	0.400 (0.015)	0.965 (0.007)	0.500 (0.023)
FCGL, unconstrained	0.0058 (0.0002)	0.425 (0.008)	0.967 (0.005)	0.440 (0.014)
Cooperative Lasso	0.0063 (0.0003)	0.440 (0.036)	0.953 (0.011)	0.471 (0.041)
UniLasso	0.0064 (0.0003)	0.642 (0.015)	0.900 (0.009)	0.171 (0.010)
HCGL, known/exclude	0.0039 (0.0002)	0.806 (0.028)	0.972 (0.009)	0.089 (0.015)

Support and sign results are not equivalent to prediction gains. Baseline exclusion reduces null selection but also loses active coefficients (Table S3). All methods have zero observed active-sign flips in that baseline sample; this is not a guarantee of correct signs. In the correlated case, the excluded HCGL fit has zero observed flips while 25.6% of active coefficients are estimated as zero and its gate assigns an incorrect direction to 12.8% of active cells. A low fitted-flip rate can therefore coexist with incorrect imposed signs.

We evaluate support at $10^{-8}, 10^{-6}$ and 10^{-4} in standardized coefficient units. Changing the originally recorded 10^{-6} classification to the displayed 10^{-4} changes mean active-sign recovery by as much as 5.00 percentage points and the selected-set false-discovery proportion by as much as 6.22 points across the reported cases. These results are tolerance-sensitive; prediction and unthresholded coefficient MSE are unaffected. Full per-seed metrics and sensitivity summaries are retained in the run outputs; Section S4 reports the other five scenarios.

These are small-dimensional exploratory experiments. Zero penalty is selected once among 120 estimated pooled-exclusion fits and twice in the known-group diagnostic; UniLasso selects its nonnegative least-squares endpoint in 56 of 120 fits. Release selects the infinite-threshold endpoint

once. These are natural domain endpoints, not evidence that the grid should extend to negative penalties or thresholds. The fixed mixing/cut settings, finite interior spacing and finite replication count still preclude globally optimized comparisons. Section S12 reports separately measured path costs; those timings are not the cost of this expanded validation search.

S4 Additional fitted-estimator summaries

These tables complete the coefficient summaries for the six-design experiment in Section 4.1. Entries are means with Monte Carlo standard errors over 20 seeds; the primary standardized selection tolerance is 10^{-4}, chosen after the independent solver audit. Recovery and flips use true active cells; FDP uses selected cells, with zero assigned to an empty selected set. Coefficient MSE uses all cells without thresholding. The known-group row is an oracle diagnostic. Numerical tolerance sensitivity is separate from Monte Carlo uncertainty.
Table S4 gives the corresponding numerical results.

Table S4: Weak signal: fitted-coefficient results, mean (Monte Carlo standard error).

Method	Coefficient MSE	Sign recovery	Sign flips	FDP
Lasso	0.0058 (0.0002)	0.5833 (0.0480)	0.0042 (0.0023)	0.4500 (0.0380)
HCGL, unconstrained	0.0058 (0.0002)	0.5792 (0.0474)	0.0042 (0.0023)	0.4580 (0.0387)
HCGL, singleton/exclude	0.0059 (0.0002)	0.5500 (0.0510)	0.0042 (0.0023)	0.4262 (0.0441)
HCGL, pooled/exclude	0.0059 (0.0002)	0.5500 (0.0510)	0.0042 (0.0023)	0.4262 (0.0441)
HCGL, pooled/release	0.0058 (0.0002)	0.5819 (0.0475)	0.0042 (0.0023)	0.4561 (0.0386)
FCGL, unconstrained	0.0062 (0.0002)	0.4792 (0.0432)	0.0056 (0.0025)	0.3781 (0.0385)
Cooperative Lasso	0.0058 (0.0002)	0.5875 (0.0474)	0.0042 (0.0023)	0.5034 (0.0395)
UniLasso	0.0069 (0.0002)	0.3236 (0.0489)	0.0014 (0.0014)	0.2298 (0.0396)
HCGL, known/exclude	0.0047 (0.0002)	0.7278 (0.0413)	0.0000 (0.0000)	0.2852 (0.0344)

Table S5 gives the corresponding numerical results.

Table S5: Partial supports: fitted-coefficient results, mean (Monte Carlo standard error).

Method	Coefficient MSE	Sign recovery	Sign flips	FDP
Lasso	0.0071 (0.0005)	0.9472 (0.0161)	0.0000 (0.0000)	0.5515 (0.0348)
HCGL, unconstrained	0.0067 (0.0004)	0.9681 (0.0123)	0.0000 (0.0000)	0.6047 (0.0217)
HCGL, singleton/exclude	0.0065 (0.0003)	0.9056 (0.0150)	0.0000 (0.0000)	0.4670 (0.0298)
HCGL, pooled/exclude	0.0065 (0.0003)	0.9097 (0.0153)	0.0000 (0.0000)	0.4737 (0.0287)
HCGL, pooled/release	0.0067 (0.0004)	0.9667 (0.0122)	0.0000 (0.0000)	0.6016 (0.0216)
FCGL, unconstrained	0.0062 (0.0002)	0.9694 (0.0060)	0.0000 (0.0000)	0.5767 (0.0087)
Cooperative Lasso	0.0071 (0.0005)	0.9486 (0.0156)	0.0000 (0.0000)	0.5524 (0.0351)
UniLasso	0.0070 (0.0003)	0.8861 (0.0110)	0.0000 (0.0000)	0.3714 (0.0133)
HCGL, known/exclude	0.0069 (0.0003)	0.9139 (0.0196)	0.0014 (0.0014)	0.4464 (0.0339)

Table S6 gives the corresponding numerical results.
Table S7 gives the corresponding numerical results.
Table S8 gives the corresponding numerical results.

Table S6: Mixed signs: fitted-coefficient results, mean (Monte Carlo standard error).

Method	Coefficient MSE	Sign recovery	Sign flips	FDP
Lasso	0.0069 (0.0004)	0.9458 (0.0130)	0.0014 (0.0014)	0.5379 (0.0322)
HCGL, unconstrained	0.0062 (0.0002)	0.9750 (0.0045)	0.0014 (0.0014)	0.6228 (0.0043)
HCGL, singleton/exclude	0.0062 (0.0003)	0.9361 (0.0119)	0.0014 (0.0014)	0.5248 (0.0244)
HCGL, pooled/exclude	0.0062 (0.0003)	0.9361 (0.0119)	0.0014 (0.0014)	0.5244 (0.0244)
HCGL, pooled/release	0.0061 (0.0002)	0.9736 (0.0047)	0.0014 (0.0014)	0.6182 (0.0046)
FCGL, unconstrained	0.0060 (0.0002)	0.9681 (0.0061)	0.0000 (0.0000)	0.5713 (0.0061)
Cooperative Lasso	0.0069 (0.0004)	0.9458 (0.0130)	0.0014 (0.0014)	0.5398 (0.0324)
UniLasso	0.0066 (0.0003)	0.8847 (0.0091)	0.0000 (0.0000)	0.3564 (0.0140)
HCGL, known/exclude	0.0099 (0.0003)	0.7569 (0.0151)	0.0000 (0.0000)	0.3857 (0.0577)

Table S7: Correlated predictors and errors: fitted-coefficient results, mean (Monte Carlo standard error).

Method	Coefficient MSE	Sign recovery	Sign flips	FDP
Lasso	0.0113 (0.0009)	0.8764 (0.0176)	0.0000 (0.0000)	0.5447 (0.0135)
HCGL, unconstrained	0.0131 (0.0011)	0.9042 (0.0154)	0.0014 (0.0014)	0.5987 (0.0165)
HCGL, singleton/exclude	0.0150 (0.0015)	0.7611 (0.0149)	0.0000 (0.0000)	0.5147 (0.0152)
HCGL, pooled/exclude	0.0163 (0.0019)	0.7444 (0.0274)	0.0000 (0.0000)	0.4944 (0.0165)
HCGL, pooled/release	0.0129 (0.0012)	0.8569 (0.0176)	0.0000 (0.0000)	0.5474 (0.0175)
FCGL, unconstrained	0.0121 (0.0011)	0.8972 (0.0172)	0.0028 (0.0019)	0.5270 (0.0343)
Cooperative Lasso	0.0101 (0.0008)	0.9389 (0.0163)	0.0000 (0.0000)	0.5234 (0.0298)
UniLasso	0.0185 (0.0017)	0.6375 (0.0211)	0.0000 (0.0000)	0.3518 (0.0255)
HCGL, known/exclude	0.0163 (0.0019)	0.7444 (0.0274)	0.0000 (0.0000)	0.4944 (0.0165)

Table S8: Unequal groups and variances: fitted-coefficient results, mean (Monte Carlo standard error).

Method	Coefficient MSE	Sign recovery	Sign flips	FDP
Lasso	0.0070 (0.0004)	0.8458 (0.0255)	0.0042 (0.0023)	0.4733 (0.0407)
HCGL, unconstrained	0.0059 (0.0003)	0.9194 (0.0083)	0.0097 (0.0042)	0.6208 (0.0067)
HCGL, singleton/exclude	0.0061 (0.0003)	0.8181 (0.0245)	0.0028 (0.0019)	0.4570 (0.0342)
HCGL, pooled/exclude	0.0059 (0.0003)	0.8306 (0.0219)	0.0028 (0.0019)	0.4652 (0.0278)
HCGL, pooled/release	0.0059 (0.0003)	0.9194 (0.0083)	0.0097 (0.0042)	0.6162 (0.0064)
FCGL, unconstrained	0.0057 (0.0003)	0.8958 (0.0108)	0.0069 (0.0028)	0.5649 (0.0090)
Cooperative Lasso	0.0070 (0.0004)	0.8542 (0.0259)	0.0056 (0.0032)	0.4776 (0.0404)
UniLasso	0.0061 (0.0004)	0.7861 (0.0207)	0.0014 (0.0014)	0.3607 (0.0201)
HCGL, known/exclude	0.0039 (0.0003)	0.9194 (0.0166)	0.0000 (0.0000)	0.2086 (0.0410)

S5 Full learned-group design and tuning protocol

The preceding low- and moderate-signal designs often produce singleton groups. We therefore prespecify a separate diagnostic in which grouping can be recovered, while retaining violations of shared support and sign. Each of five designs has 40 paired seeds (1000–1039), 80 training observations, 1,000 independent test observations, 12 predictors and three response blocks of size four. The population signal share is 0.65 for each response, including the covariance contribution in the correlated-predictor design. This is a favorable grouped-signal regime, not an estimate of signal strength in the yeast application.

Each block has two shared anchor predictors and a weaker third predictor, with base magnitudes $(1, 1, 0.6)$, independent block and member multipliers uniform on $[0.9, 1.1]$, and shared randomly assigned directions. Baseline supports are disjoint across blocks. The mixed-direction design reverses the third sign for alternate members. The partial-support design instead replaces that predictor by a block-specific unused predictor for alternate members. Two separate stress designs retain coherent supports/signs and introduce either predictor covariance $\Sigma_{ij} = 0.6^{|i-j|}$ or within-block error correlation 0.6. Parameter, training and test streams are independent; no truth enters fitting.

We compare the eight feasible methods from Section 4.1. The initial three-fold inner validation selects every applicable combination of cut $\kappa \in \{0.35, 0.5, 0.7\}$, mixing $\alpha \in \{0.25, 0.5, 0.75\}$, gate threshold $\tau \in \{0, 1.5, 3\}$, and the stated penalty fractions. Lasso and UniLasso tune only their penalty; cooperative Lasso also tunes the cut; unconstrained HCGL/FCGL tune cut and mixing. Every candidate uses preprocessing estimated within its inner training fold. Known labels are used only to score partition recovery. All fits have zero targets and nonadaptive weights. The initial pooled-exclusion fits select the upper mixing value 0.75 in 92.5–100% of replications. After inspecting this boundary diagnostic, we extend the mixing grid to include $\alpha = 1$ for every applicable method, recompute its cut/threshold/penalty candidates on the same inner folds, and select over the combined candidate set. A further boundary audit extends the cut grid to $\{0, 0.175, 0.35, 0.5, 0.7, 0.85, 1\}$, mixing to $\{0, 0.1, 0.25, 0.5, 0.75, 1\}$, and signed-method thresholds to $\{0, 0.75, 1.5, 3, 6, \infty\}$. Every method receives the fourteen penalty fractions in Section 4.1, including zero. The cut endpoints give singleton and fully pooled groups, and $\tau = \infty$ denotes complete gate failure. Mixing zero uses the appropriate pure-group dual-norm penalty scale; zero penalty uses constrained least squares. All jointly tuned results below use this final grid, reaching the natural parameter-domain endpoints. Both extensions followed inspection of the earlier selection counts and were not part of the initial protocol. The original results remain archived. Test outcomes and coefficient truth never select a candidate. Earlier CV cells are retained after source, input and output-hash checks; added cells use the same three inner folds. A fresh-CV reconstruction separately checks the earlier reused losses. Finite grid spacing and CV sampling variability remain.

A paired reference holds $(\kappa, \alpha, \tau) = (0.5, 0.5, 1.5)$ and tunes only the penalty over the same final fourteen fractions; one-factor sensitivity changes one of those settings while holding the other two fixed and reselecting the penalty. These comparisons isolate changes in the gate or penalty more directly than comparing separately joint-tuned procedures. We report prediction, coefficient error, fitted signs and support, and the actual grouping rather than assuming that latent groups were recovered.

Each jointly tuned procedure may select a different partition. The following intervals describe paired Monte Carlo uncertainty and are not adjusted for multiple comparisons or the disclosed post-inspection grid extensions.
Table S9 gives the corresponding numerical results.

Table S9: Grouping diagnostics at the fixed reference cut and paired prediction differences for jointly tuned procedures. ARI compares learned groups with the designed anchor partition. Opposite active pairs is the fraction of within-learned-group response–predictor pairs with two nonzero coefficients of opposite signs. The final column is pooled-exclusion HCGL minus unconstrained HCGL standardized test MSE, with a paired t_{39} interval; its independently tuned partitions need not match those underlying the first three columns.

Design	Mean groups	ARI	Opposite active pairs (%)	Joint-CV ΔMSE [95% CI]
Coherent	3.000	1.000	0.00	-0.0129 [-0.0156, -0.0101]
Mixed directions	5.075	0.617	9.92	-0.0056 [-0.0079, -0.0033]
Partial support	3.225	0.963	0.00	-0.0049 [-0.0073, -0.0025]
Predictor correlation	3.000	1.000	0.00	-0.0007 [-0.0099, 0.0086]
Residual correlation	3.000	1.000	0.00	-0.0054 [-0.0086, -0.0021]

S6 Additional learned-group results

The following tables score the jointly tuned fitted coefficients at the common standardized coefficient tolerance 10^{-4}. The stored original 10^{-6} classification is more sensitive to solver residuals; the displayed threshold was chosen after the independent numerical audit, uniformly across methods. All cells are means (Monte Carlo standard errors) over 40 seeds. Coefficient MSE is in training-standardized units and multiplied by 10^3; support and active-sign quantities are percentages. FDP uses selected coefficients as its denominator; correct/flip/abstention partition the truly active cells. Full precision and recall summaries are saved with the results. The empty-selected-set FDP convention is zero.
Table S10 gives the corresponding numerical results.

Table S10: Coherent: jointly tuned coefficient and support results.

Method	10^3 MSE	FDP	Correct	Flip	Abstain
Lasso	3.03 (0.12)	47.81 (0.88)	99.86 (0.10)	0.00 (0.00)	0.14 (0.10)
HCGL, unconstrained	3.03 (0.11)	48.07 (0.89)	99.86 (0.10)	0.00 (0.00)	0.14 (0.10)
HCGL, singleton/exclude	2.74 (0.13)	38.18 (2.15)	99.24 (0.24)	0.00 (0.00)	0.76 (0.24)
HCGL, pooled/exclude	1.97 (0.13)	25.68 (2.44)	99.38 (0.39)	0.00 (0.00)	0.62 (0.39)
HCGL, pooled/release	2.77 (0.12)	45.73 (1.66)	99.65 (0.28)	0.00 (0.00)	0.35 (0.28)
FCGL, unconstrained	2.24 (0.09)	57.96 (1.02)	100.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Cooperative Lasso	2.16 (0.09)	41.04 (1.35)	100.00 (0.00)	0.00 (0.00)	0.00 (0.00)
UniLasso	2.23 (0.10)	29.10 (1.26)	99.58 (0.19)	0.00 (0.00)	0.42 (0.19)

Table S11 gives the corresponding numerical results.

Table S11: Mixed directions: jointly tuned coefficient and support results.

Method	10^3 MSE	FDP	Correct	Flip	Abstain
Lasso	3.02 (0.11)	46.70 (0.71)	99.93 (0.07)	0.00 (0.00)	0.07 (0.07)
HCGL, unconstrained	3.04 (0.12)	47.48 (0.93)	99.93 (0.07)	0.00 (0.00)	0.07 (0.07)
HCGL, singleton/exclude	2.83 (0.14)	39.57 (1.90)	99.10 (0.38)	0.00 (0.00)	0.90 (0.38)
HCGL, pooled/exclude	2.58 (0.14)	34.80 (2.18)	98.96 (0.39)	0.00 (0.00)	1.04 (0.39)
HCGL, pooled/release	3.09 (0.14)	43.18 (0.84)	99.38 (0.27)	0.00 (0.00)	0.62 (0.27)
FCGL, unconstrained	2.25 (0.09)	58.90 (0.96)	100.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Cooperative Lasso	2.48 (0.10)	42.25 (1.23)	100.00 (0.00)	0.00 (0.00)	0.00 (0.00)
UniLasso	2.27 (0.10)	29.00 (1.34)	99.51 (0.22)	0.00 (0.00)	0.49 (0.22)

Table S12 gives the corresponding numerical results.
Table S13 gives the corresponding numerical results.

Table S12: Partial support: jointly tuned coefficient and support results.

Method	10^3 MSE	FDP	Correct	Flip	Abstain
Lasso	3.02 (0.12)	47.83 (0.83)	99.93 (0.07)	0.00 (0.00)	0.07 (0.07)
HCGL, unconstrained	3.01 (0.12)	48.07 (0.84)	99.93 (0.07)	0.00 (0.00)	0.07 (0.07)
HCGL, singleton/exclude	2.75 (0.14)	38.06 (2.18)	99.44 (0.23)	0.00 (0.00)	0.56 (0.23)
HCGL, pooled/exclude	2.60 (0.14)	38.17 (2.52)	99.10 (0.29)	0.00 (0.00)	0.90 (0.29)
HCGL, pooled/release	2.84 (0.11)	44.59 (1.38)	99.79 (0.15)	0.00 (0.00)	0.21 (0.15)
FCGL, unconstrained	2.70 (0.11)	61.72 (0.95)	100.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Cooperative Lasso	2.59 (0.11)	46.96 (1.36)	100.00 (0.00)	0.00 (0.00)	0.00 (0.00)
UniLasso	2.18 (0.10)	28.39 (1.34)	99.79 (0.15)	0.00 (0.00)	0.21 (0.15)

Table S13: Predictor correlation: jointly tuned coefficient and support results.

Method	10^3 MSE	FDP	Correct	Flip	Abstain
Lasso	5.20 (0.31)	48.61 (1.86)	99.72 (0.13)	0.00 (0.00)	0.28 (0.13)
HCGL, unconstrained	5.25 (0.31)	50.25 (1.95)	99.72 (0.13)	0.00 (0.00)	0.28 (0.13)
HCGL, singleton/exclude	6.87 (0.98)	48.59 (1.13)	96.25 (0.88)	0.00 (0.00)	3.75 (0.88)
HCGL, pooled/exclude	6.02 (0.98)	47.93 (0.90)	97.43 (0.76)	0.00 (0.00)	2.57 (0.76)
HCGL, pooled/release	4.70 (0.40)	51.16 (0.97)	99.03 (0.47)	0.00 (0.00)	0.97 (0.47)
FCGL, unconstrained	3.54 (0.23)	57.02 (1.40)	100.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Cooperative Lasso	3.76 (0.31)	43.27 (2.27)	100.00 (0.00)	0.00 (0.00)	0.00 (0.00)
UniLasso	7.29 (1.15)	38.59 (1.01)	95.28 (0.99)	0.00 (0.00)	4.72 (0.99)

Table S14 gives the corresponding numerical results.

Table S14: Residual correlation: jointly tuned coefficient and support results.

Method	10^3 MSE	FDP	Correct	Flip	Abstain
Lasso	2.87 (0.16)	46.24 (1.07)	100.00 (0.00)	0.00 (0.00)	0.00 (0.00)
HCGL, unconstrained	2.87 (0.16)	46.56 (1.08)	100.00 (0.00)	0.00 (0.00)	0.00 (0.00)
HCGL, singleton/exclude	2.65 (0.18)	33.96 (2.52)	98.89 (0.42)	0.00 (0.00)	1.11 (0.42)
HCGL, pooled/exclude	2.45 (0.18)	31.23 (2.38)	99.38 (0.39)	0.00 (0.00)	0.62 (0.39)
HCGL, pooled/release	2.75 (0.16)	42.54 (1.65)	100.00 (0.00)	0.00 (0.00)	0.00 (0.00)
FCGL, unconstrained	2.46 (0.14)	55.54 (1.21)	100.00 (0.00)	0.00 (0.00)	0.00 (0.00)
Cooperative Lasso	2.45 (0.14)	48.92 (1.25)	100.00 (0.00)	0.00 (0.00)	0.00 (0.00)
UniLasso	2.38 (0.19)	26.92 (1.47)	99.03 (0.38)	0.00 (0.00)	0.97 (0.38)

Comparing the recorded 10^{-6} classification with displayed 10^{-4} changes mean active-sign recovery by at most 0.139 percentage points among the jointly tuned fits, but changes mean FDP by as much as 7.13 points. Prediction and unthresholded coefficient MSE are unchanged. Thus support-count and FDP rankings still require the declared numerical convention; tolerance summaries are supplied for every method, design and sensitivity variant. At $\alpha = 1$, the two unconstrained conic objectives coincide with ordinary Lasso. A separate numerical diagnostic uses the earlier seven-positive-fraction grid and fixed cut 0.5, independently of the final endpoint-grid comparisons. Across all 200 datasets the conic implementations choose the same penalty as the coordinate-descent implementation, with maximum test-MSE difference 9.91×10^{-6}. Nevertheless, their mean FDP difference is 4.30 percentage points at 10^{-6}, falling to 0.300 points at 10^{-4} (maximum 2.37 points). Active-sign recovery and support recall agree exactly at 10^{-4}. These are numerical classification effects, not an advantage of one identical objective.

S6.1 Paired tuning sensitivity

Each one-factor change below is compared on the same 40 datasets with the reference cut 0.5, mixing 0.5 and threshold 1.5. Every fixed-setting fit tunes only its penalty fraction over the final fourteen-value grid, including zero. The joint-CV row uses the same penalty grid and selects all applicable settings within the training folds, including both mixing and cut endpoints and complete gate failure. Entries are paired changes in standardized test MSE for pooled-exclusion HCGL, with t_{39} intervals; positive changes are worse. Corresponding all-method contrasts, all added one-factor settings and the full CV surfaces are retained in the replication outputs. The table shows selected neighboring settings.
Table S15 gives the corresponding numerical results.

Table S15: Selected paired one-factor and joint-tuning sensitivity for pooled exclusion. All rows and their reference use the extended penalty grid.

Design	Change	ΔMSE [95% CI]
Coherent	Cut 0.35	0.0033 [0.0013, 0.0053]
	Cut 0.70	0.0000 [0.0000, 0.0000]
	$\alpha = 0.25$	0.0000 [0.0000, 0.0001]
	$\alpha = 0.75$	-0.0001 [-0.0002, 0.0001]
	$\tau = 0$	0.0137 [0.0107, 0.0167]
	$\tau = 3$	0.0254 [0.0171, 0.0337]
	Joint CV	0.0015 [0.0000, 0.0030]
Mixed directions	Cut 0.35	-0.0187 [-0.0266, -0.0109]
	Cut 0.70	0.0683 [0.0577, 0.0789]
	$\alpha = 0.25$	0.0002 [0.0000, 0.0003]
	$\alpha = 0.75$	-0.0001 [-0.0003, 0.0000]
	$\tau = 0$	0.0021 [-0.0020, 0.0063]
	$\tau = 3$	0.0162 [0.0099, 0.0224]
	Joint CV	-0.0183 [-0.0262, -0.0105]
Partial support	Cut 0.35	-0.0322 [-0.0387, -0.0256]
	Cut 0.70	0.0044 [0.0007, 0.0080]
	$\alpha = 0.25$	0.0002 [0.0000, 0.0003]
	$\alpha = 0.75$	-0.0002 [-0.0004, 0.0001]
	$\tau = 0$	-0.0232 [-0.0294, -0.0169]
	$\tau = 3$	0.0379 [0.0294, 0.0465]
	Joint CV	-0.0303 [-0.0365, -0.0240]
Predictor correlation	Cut 0.35	0.0009 [-0.0044, 0.0063]
	Cut 0.70	0.0055 [-0.0021, 0.0131]
	$\alpha = 0.25$	0.0000 [-0.0002, 0.0003]
	$\alpha = 0.75$	-0.0006 [-0.0011, -0.0002]
	$\tau = 0$	-0.0098 [-0.0211, 0.0015]
	$\tau = 3$	0.0376 [0.0174, 0.0578]
	Joint CV	-0.0158 [-0.0268, -0.0047]
Residual correlation	Cut 0.35	0.0000 [0.0000, 0.0000]
	Cut 0.70	0.0000 [0.0000, 0.0000]
	$\alpha = 0.25$	-0.0002 [-0.0005, 0.0001]
	$\alpha = 0.75$	-0.0004 [-0.0008, 0.0000]
	$\tau = 0$	0.0133 [0.0080, 0.0186]
	$\tau = 3$	0.0431 [0.0313, 0.0550]
	Joint CV	0.0021 [-0.0007, 0.0049]

Table S16 gives the corresponding numerical results.

The initial upper mixing endpoint 0.75 was selected in 92.5–100% of pooled-exclusion fits, prompting the first extension to 1. The final extension also covers the other natural endpoints.

Table S16: Joint-CV selections for pooled-exclusion HCGL, counts out of 40. Each block reports its own parameter marginal; these are not joint cell counts. At $\alpha = 1$, distinct cuts can produce the same gate and fitted coefficients; their counts identify a retained representative rather than a unique partition.

Cut κ	0	0.175	0.35	0.5	0.7	0.85	1
Coherent	0	0	4	28	0	6	2
Mixed directions	8	0	26	6	0	0	0
Partial support	11	0	21	3	2	2	1
Predictor correlation	3	0	9	21	2	3	2
Residual correlation	1	6	19	0	0	4	10
Mixing α	0	0.1	0.25	0.5	0.75	1	
Coherent	3	1	1	0	3	32	
Mixed directions	0	0	0	0	0	40	
Partial support	0	0	0	0	1	39	
Predictor correlation	0	0	0	1	0	39	
Residual correlation	3	0	0	0	0	37	
Gate τ	0	0.75	1.5	3	6	∞	
Coherent	2	9	28	1	0	0	
Mixed directions	6	15	19	0	0	0	
Partial support	13	10	17	0	0	0	
Predictor correlation	18	11	10	1	0	0	
Residual correlation	13	6	21	0	0	0	

Table S16 gives all final cut, mixing and threshold counts. Pure L1 remains common, but pure-group mixing is selected in three coherent and three residual-correlation replications. At $\alpha = 1$, the unconstrained HCGL/FCGL objectives coincide, and their cut has no objective meaning. At zero penalty the mixing value has no effect. Different cuts can produce identical partitions and losses. The stored parameter-candidate order is used first, followed by descending penalty fractions within each candidate; the first minimum wins.

Pooled exclusion chooses zero penalty in one coherent replication and positive fractions below the earlier 0.001 floor in 7, 1, 0, 0 and 3 replications in the coherent, mixed, partial, predictor- and residual-correlation designs. Across all methods, UniLasso chooses its zero-penalty second-stage endpoint in 40 of 200 replications. Seven pooled-exclusion fits choose the smallest positive fraction 10^{-6} although zero is available. Holding their other settings fixed, their test MSE differs from the zero-penalty fit by at most 6.65×10^{-6}. Across both synthetic studies, eight release-on-failure fits select $\tau = 6$; their CV losses differ from complete release at $\tau = \infty$ by at most 1.47×10^{-6}, including four exact ties. Such differences give little evidence for a well-separated tuning optimum. All-method endpoint counts and earlier-to-final contrasts are saved. Natural endpoint coverage removes exterior truncation, while finite spacing and CV variability remain. The final joint-CV procedure changes residual-correlation MSE relative to the fixed reference by 0.00209 (95% interval $[-0.00070, 0.00488]$).

S7 Full application sensitivity settings

Ten one-factor neighbors vary LD pruning $(0.85, 0.98)$, outcome-screening p $(0.005, 0.02)$, Ward cut height as a fraction of the final merge height $(0.5, 0.9)$, gate threshold $(1, 3)$, or HCGL mixing weight $(0.2, 0.8)$. The baseline is $(0.92, 0.01, 0.7, 2, 0.5)$, respectively; screening requires at least two associated genes. All settings retain the same three repeats, three outer folds and three inner folds. All preprocessing is fitted separately on each inner or outer training fold, and only the penalty fraction is chosen by inner validation. These are fixed sensitivity settings, with no setting selected

from outer outcomes. Table S17 gives the corresponding numerical results.

Table S17: Prespecified eQTL sensitivity. The first four numeric columns are repeat-mean median-gene R^2; the last is repeat-mean pooled R^2 for exclusion. U_H, E_H and R_H denote unconstrained, pooled-exclusion and release-on-failure HCGL.

Setting	Lasso	U_H	E_H	R_H	E_H pooled
Baseline	0.051	0.073	0.035	0.073	0.398
LD 0.85	0.053	0.073	0.028	0.072	0.375
LD 0.98	0.050	0.074	0.034	0.074	0.425
Screen $p = 0.005$	0.050	0.070	0.033	0.070	0.398
Screen $p = 0.02$	0.055	0.079	0.035	0.079	0.397
Cut 0.5	0.051	0.063	0.043	0.063	0.521
Cut 0.9	0.051	0.085	0.026	0.084	0.309
$\tau = 1$	0.051	0.073	0.049	0.070	0.405
$\tau = 3$	0.051	0.073	0.004	0.073	0.355
$\alpha = 0.2$	0.051	0.082	0.037	0.081	0.362
$\alpha = 0.8$	0.051	0.064	0.033	0.064	0.425

Pooled exclusion has lower median-gene R^2 than both matched unconstrained and release-on-failure HCGL in every repeat at every reported setting. Its mean paired loss against unconstrained HCGL ranges from 0.020 to 0.069. The size of the loss is sensitive to the cut and gate: excluded pooled R^2 is 0.521 at cut 0.5 and 0.309 at cut 0.9, while excluded median-gene R^2 is 0.049 at $\tau = 1$ and 0.004 at $\tau = 3$. Release remains close to unconstrained HCGL in median-gene prediction (mean paired differences -0.0026 to 0.0004), without establishing equivalence. None of the outer fits selected either penalty-grid endpoint. These results support a descriptive sensitivity conclusion on this panel, rather than a validated optimum over settings. Section S8 reports paired split dispersions and same-fold changes in partitions, markers and fitted selection, together with the coefficient-threshold audit.

S8 Additional eQTL sensitivity results

Table S18 gives the corresponding numerical results.

Table S18: Paired differences in median-gene R^2 over the same three repeats. Entries are mean (split SD), not standard errors or confidence intervals.

Setting	$E_H - U_H$	$E_H - R_H$
Baseline	-0.038 (0.005)	-0.038 (0.007)
LD 0.85	-0.045 (0.009)	-0.044 (0.009)
LD 0.98	-0.041 (0.003)	-0.040 (0.002)
Screen $p = 0.005$	-0.037 (0.005)	-0.037 (0.005)
Screen $p = 0.02$	-0.044 (0.002)	-0.044 (0.004)
Cut 0.5	-0.020 (0.004)	-0.020 (0.004)
Cut 0.9	-0.059 (0.012)	-0.058 (0.013)
$\tau = 1$	-0.024 (0.009)	-0.021 (0.011)
$\tau = 3$	-0.069 (0.004)	-0.068 (0.003)
$\alpha = 0.2$	-0.045 (0.006)	-0.044 (0.006)
$\alpha = 0.8$	-0.032 (0.005)	-0.032 (0.005)

Table S19 gives the corresponding numerical results.

The two cut neighbors produce 6–11 and 2–3 clusters, compared with 3–4 at baseline. Marker selection and fitted support change materially with LD pruning, screening, the cut, and the gate. For pooled exclusion, same-fold selection Jaccard is 0.388 and 0.423 under the LD neighbors; restricting its denominator to markers retained under both settings increases these values to 0.791

Table S19: Same-fold changes relative to baseline, averaged over nine paired training splits. ARI compares gene partitions; marker J compares retained marker sets; the last four columns are fitted-selection Jaccard indices. An absent marker is distinct from a retained zero coefficient. Selection uses $|\widehat{B}| > 10^{-6}$ in training-standardized units. These comparisons are distinct from between-split stability.

Setting	ARI	Marker J	Lasso	U_H	E_H	R_H
Baseline	1.000	1.000	1.000	1.000	1.000	1.000
LD 0.85	1.000	0.470	0.362	0.407	0.388	0.411
LD 0.98	1.000	0.491	0.384	0.432	0.423	0.426
Screen $p = 0.005$	1.000	0.600	0.679	0.570	0.708	0.573
Screen $p = 0.02$	1.000	0.605	0.762	0.593	0.754	0.587
Cut 0.5	0.491	1.000	1.000	0.552	0.469	0.545
Cut 0.9	0.713	1.000	1.000	0.723	0.593	0.717
$\tau = 1$	1.000	1.000	1.000	1.000	0.460	0.927
$\tau = 3$	1.000	1.000	1.000	1.000	0.234	0.958
$\alpha = 0.2$	1.000	1.000	1.000	0.433	0.656	0.456
$\alpha = 0.8$	1.000	1.000	1.000	0.390	0.529	0.387

and 0.781. All jointly selected cells keep their direction in same-fold comparisons, but this conditional agreement does not establish stable overall support or correct partial-effect signs. A post hoc audit of saved coefficients changes the selection threshold from 10^{-6} to 10^{-4} without refitting. Across all methods and settings, mean between-split selection Jaccard changes by at most 0.0034, common-marker Jaccard by at most 0.0102, and conditional direction agreement by at most 0.0073. At 10^{-8} the corresponding maximal changes are 0.0550, 0.2773, and 0.1053, demonstrating sensitivity to small numerical coefficients. The primary threshold remains the prespecified 10^{-6}. Mean between-split conditional direction agreement across the sensitivity settings ranges from 0.963 to 1.000; the higher baseline values should not be generalized to every setting.

S9 Joint tuning of the eQTL protocol

The jointly tuned protocol of the main text uses the repeats, outer folds and inner folds of the fixed-setting protocol (seed 20261024, three by three by three), the same training-only preprocessing chain and the same eight estimators. Within each inner fold the preprocessing is fitted once, and the clusters and gate signs are recomputed for every cut and threshold on the grid. For each method the candidate with the smallest pooled inner standardized squared error over its own grid is refitted on the outer training set. Lasso tunes the fraction only. Singleton gating tunes the threshold and the fraction. The unconstrained HCGL and FCGL fits tune the cut and the fraction, because the cut enters the HCGL group multiplier and the FCGL block structure. The pooled and singleton-gated HCGL and FCGL fits tune all three. Ties choose the earliest candidate in grid order with the larger fraction. The producer is `eqtl_joint_tuning.py`, which also runs the fixed-setting arm, and `eqtl_joint_report.py` writes the tables.

The search ran in a Linux container with the `factorlasso` 1.0.5 wheel from PyPI and the CLARABEL solver, in two processes with per-fold and per-inner-fold checkpoints, with about 16 processor hours of solver time in the inner loops of the joint arm (11,340 candidate fits). The fixed-setting arm under CLARABEL reproduces all 24 summary values of the archived MOSEK run (median, mean and pooled R^2 for eight methods) with a largest absolute difference of 7×10^{-6}, and selects fraction 0.3 in all 72 outer fits as before. CLARABEL reported a possibly inaccurate solution for 70 of the roughly 12,000 solves of both arms. Those solves were not mapped to candidates. Every selected candidate uses fraction 0.17 or 0.3.

Table S20 lists the selected settings over the nine outer fits, and Table S21 the paired median-gene contrasts for both arms.

Pooled exclusion selects the finest cut and the lowest threshold on the grid in every fit, so

Table S20: Selected cut, threshold and penalty fraction over the nine outer fits of the jointly tuned eQTL protocol. Entries are value: count.

Method	Cut κ	Threshold τ	Fraction of $\lambda_{\max}$
Lasso	–	–	0.3: 9
Singleton-gated Lasso	–	1: 3, 2: 6	0.3: 9
HCGL, unconstrained	0.35: 1, 0.5: 3, 0.7: 5	–	0.3: 9
HCGL, singleton/exclude	0.35: 2, 0.5: 2, 0.7: 5	1: 8, 2: 1	0.3: 9
HCGL, pooled/exclude	0.35: 9	1: 9	0.3: 9
HCGL, pooled/release	0.35: 1, 0.5: 3, 0.7: 5	1: 3, 2: 2, 3: 4	0.3: 9
FCGL, unconstrained	0.5: 3, 0.7: 1, 0.9: 5	–	0.17: 4, 0.3: 5
FCGL, pooled/exclude	0.35: 7, 0.5: 2	1: 9	0.17: 4, 0.3: 5

Table S21: Paired differences in median-gene R^2 over the same three repeats, fixed-setting arm and jointly tuned arm. Entries are mean (split SD) and the number of repeats with a negative difference.

Contrast	Fixed settings	Joint tuning
Pooled exclusion minus unconstrained HCGL	−0.0384 (0.0052), 3/3	−0.0073 (0.0119), 2/3
Pooled exclusion minus singleton exclusion HCGL	−0.0307 (0.0045), 3/3	−0.0065 (0.0111), 2/3
Pooled release minus unconstrained HCGL	−0.0001 (0.0014), 1/3	−0.0010 (0.0017), 1/3
FCGL exclusion minus unconstrained FCGL	−0.0195 (0.0068), 3/3	−0.0173 (0.0098), 3/3

its tuned form is close to singleton gating with a weak gate. The selected settings of the other methods vary across splits, which is the selection variability of a $4 \times 3 \times 7$ grid on inner folds of about 25 segregants. The joint search does not change the stability and sensitivity analyses of the preceding sections, which concern the fixed-setting protocol.

S10 Variance-convention sensitivity in existing applications

We compare the independent-cell and date-score gates in 35 paired cases drawn from existing package examples and the yeast application. Within each pair, only `auto_sign_variance` changes: training data, preprocessing, clusters, threshold, numerical penalty, prior, adaptive-weight configuration and solver are identical. Marginal slopes are unchanged in every pair. This is a fixed-setting sensitivity analysis, not a comparison of separately tuned procedures or a replacement for the nested analysis. Package defaults remain `auto_sign_constraints=False`, `auto_sign_threshold_t=0.75` and `auto_sign_variance="independent"`.

Synthetic applications. The 17 synthetic pairs retain the examples' original seeds and configurations. Credit attribution uses HCGL with zero or credit-prior centers and FCGL with credit-prior centers, each at $\lambda \in \{10^{-7}, 10^{-5}, 10^{-4}, 10^{-3}\}$, $\tau = 1$ and $\alpha = 0$. Adaptive weights have exponent one and floor 0.5. The quickstart uses $\tau = 0.75$, $\alpha = 0$ and no adaptive weights or prior; its expanding-window CV selects $\lambda = 10^{-5}$ on the relevant training sample, both for the full 120 observations and for the first 96 observations preceding a 24-observation holdout. That selected numerical penalty is then shared by the variance modes. The production teaching example uses FCGL, $\lambda = 10^{-5}$, $\tau = 0.75$, $\alpha = 0$, OLS priors and adaptive weights with exponent one and floor 10^{-3}, on 72 observations or a 54/18 split. Its fitting loss uses EWMA span 36 and weight-mass normalization; its sign statistics have equal row weights on the EWMA-demeaned panel. All other synthetic fits use equal-weight demeaning and sample-normalized loss, and all gates have equal row weights. Credit, quickstart and production retain their package clustering convention (Section S17), with cut fractions 0.4, 0.5 and 0.5, respectively. The low-signal teaching panel uses four known groups of six responses, eight predictors, $\lambda = 10^{-3}$, $\tau = 0.75$, $\alpha = 0$ and no adaptive

weights or prior. These synthetic fits use CLARABEL.

Define the relative coefficient change $d_B = \|\widehat{B}_{\text{date}} - \widehat{B}_{\text{independent}}\|_F / \|\widehat{B}_{\text{independent}}\|_F$. Table S22 reports $100d_B$ and changes in pooled R^2, using the training response mean as the prediction baseline. The four quickstart holdout gate changes are new exclusions; its assembled covariance changes by 0.348% in relative Frobenius norm. The six teaching-panel changes are newly retained cells, so date variance does not uniformly tighten the gate. Both production partitions consist of singleton responses, limiting their evidence about multiresponse pooling; only their point estimates, not bootstrap intervals, are compared. The 14 adaptive configurations have unchanged gates, slopes and weights. These are synthetic illustrations, not live portfolio backtests.

Table S22: Synthetic variance sensitivity. Gate counts are changed coefficient cells per fit; all fits within a grouped row have the same counts and displayed changes. ΔR^2 is date minus independent, in R^2 units. In-sample changes do not measure predictive improvement.

Application	Pairs	Gates	$100d_B$	ΔR^2	Evaluation
Credit	12	0/39	0.000	0.000000	In sample
Quickstart full	1	0/48	0.000	0.000000	In sample
Quickstart holdout	1	4/48	0.288	-0.000030	Held out
Production	2	0/4	0.000	0.000000	Both
Known-group teaching	1	6/192	19.955	0.006084	In sample

Yeast comparison. We reuse the nine outer training/test splits (three repeats of three folds) on 112 segregants and 64 genes. Each training set independently fits LD pruning at 0.92, screening at $p = 0.01$ for at least two genes, means and population-form standard deviations, and Ward clustering at 0.7 of the final merge height. Both gates use $\tau = 2$, $\alpha = 0.5$, equal observation weights, zero targets and no adaptive weights. For each split and penalty geometry, λ is fixed at 0.3 times the *unconstrained* training KKT zero-solution threshold and is shared by the variance modes. There is no inner selection or retuning in this additional comparison. The supplied training partition implements the HCGL row penalty through `GROUP_LASSO`; FCGL uses the same external partition. Both use normalized group weights, sample-normalized loss and MOSEK.

The reported coefficient changes are computed in each fold's training-standardized units, then summarized by the median and range over nine splits. The gate fraction divides the total number of changed cells by the total number of *all* coefficient cells across those splits, not by previously retained cells. Standardized held-out MSE pools squared errors over all repeats, held-out individuals and genes:

$$\text{SMSE}_v = \frac{1}{3 \cdot 112 \cdot 64} \sum_{r=1}^{3} \sum_{f=1}^{3} \sum_{t \in I_{rf}} \sum_{k=1}^{64} \left(\frac{y_{tk} - \widehat{y}^{(v)}_{rf,tk}}{\widehat{s}_{y,rf,k}} \right)^2, \tag{S11}$$

where I_{rf} is the held-out fold and $\widehat{s}_{y,rf,k}$ is its training standard deviation. Each segregant appears once per repeat. Repeated splits are not independent biological replications.

Table S23: Yeast variance sensitivity with a common numerical penalty within each pair. Gate changes are percentages of all coefficient cells; all are new exclusions. Coefficient changes are median (minimum, maximum) percentages over nine splits. MSE is defined in Equation (S11); the last column is $100(\text{SMSE}_{\text{date}} / \text{SMSE}_{\text{independent}} - 1)$.

Method	Gate (%)	$100d_B$	MSE indep.	MSE date	Increase (%)
HCGL	32.71	53.62 (28.30, 59.63)	0.980582	0.994230	1.39
FCGL	32.71	18.10 (3.23, 43.37)	0.974422	0.979883	0.56

The gate-change fraction ranges from 26.93% to 42.49% across splits; no yeast gate is newly retained. Mean selected-cell counts at $|\widehat{B}| > 10^{-4}$ fall from 1446.56 to 759.33 for HCGL and from

310.44 to 262.89 for FCGL. At 10^{-6} the corresponding counts are 1530.22 to 823.56 and 321.89 to 316.00, showing the relevance of numerical support thresholds. Original-expression pooled R^2, pooling errors and training-mean baseline errors across all nine splits, changes from 0.405309 to 0.397574 for HCGL and from 0.424316 to 0.418976 for FCGL. The date-score choice changes fitted selection materially and slightly worsens aggregate prediction under these fixed settings. Accounting for within-date response dependence does not guarantee predictive improvement, calibrated sign-error control or stable biological discoveries. The statistic assumes independent dates and is not HAC. Reproduction uses `benchmarks/compare_sign_variance.py`; the accompanying `verify_variance_sensitivity.py` checks these tables and summaries directly from preserved coefficient, gate and prediction arrays without refitting.

S11 Held-out marginal diagnostics

S11.1 Held-out marginal-sign diagnostics

For each outer split, we keep the training-selected markers and response clusters fixed and compute marginal association signs using only the held-out segregants. At the descriptive two-sided $p \leq 0.05$ threshold, each cluster–marker cell with at least two nonzero signs contributes the majority sign fraction. Average coherence over the nine outer splits is 0.869, compared with 0.801 for 100 random partitions per split that preserve cluster sizes. The training pooled gate and nonzero held-out marginal signs agree in 0.843 of their jointly retained cells on average across splits. These denominators select detectable marginal associations and the repeated splits overlap; the figures are descriptive stability measures without a multiplicity-adjusted significance interpretation. They support some out-of-sample marginal coherence, not validation of conditional coefficient directions, biological pathways or causal regulation.

S12 Computation and reproducibility

The manuscript distinguishes the archived independent-gate experiments, the September date-score gate validation and the revised complete-estimator experiments. The historical archive at `https://doi.org/10.5281/zenodo.21000294` pins `factorlasso` 0.7.2; its data and result caches have not been overwritten. The working source is developed at `https://github.com/ArturSepp/factorlasso/tree/main/papers/sign_pooling_2026`. New run receipts record source hashes, configuration, seeds, solver and dependency versions. The eQTL receipt also hashes its frozen inputs and saved outputs; the synthetic experiments identify generated inputs through their producer, configuration and seeds. The integration manifest hashes the final result files. A source revision and its installed distribution metadata are recorded separately when they differ. Online Resource 2 contains the exact revised source, results, analysis scripts and observed dependency versions, with entry hashes and verification receipts. The historical DOI does not contain these new results.

The complete-estimator studies, preceding the variance comparison in Section S10, were reproduced in a newly created Windows CPython 3.12.7 environment, using a freshly built `factorlasso` 1.0.5 wheel and an exported source tree outside the development checkout. The complete runtime dependency closure is pinned separately. Both final synthetic grids recomputed every validation loss without earlier fit caches; all 2,680 primary fits, 16,800 sensitivity refits and nested eQTL fits reproduced their saved coefficients and performance metrics exactly. Small differences between the staged and direct validation losses selected different representatives among equivalent pure-L1 cuts. Their objectives and constraints were checked on every inner fold and full training sample; fitted coefficients were identical. The cut-frequency table reports the fresh selections and does not imply a uniquely identified partition. The timing table also reports the fresh run. Independent table checks verify 1,235 synthetic and 245 eQTL printed values. This verifies a clean Python

installation on the same host; PDF rebuilding uses the existing TeX installation, with its template dependencies recorded.

The later variance comparison uses a source checkout declaring version 1.0.5, with the subsequent package gate-endpoint repair. Its receipts identify the imported source files separately from installed distribution metadata. Online Resource 2 preserves this later source and evidence alongside the original clean-reproduction baseline; the earlier clean-installation claim does not extend to this additional run.

The revised eQTL command is `eqtl_validation.py`. It starts from the raw frozen marker panel and repeats LD thinning, outcome screening, scaling, clustering and sign derivation inside every relevant training fold. Its outputs include fold manifests, selected penalties and markers, coefficients, raw out-of-fold predictions, split summaries and held-out diagnostics. `estimator_benchmark.py` implements the paired synthetic comparison with independent test samples and feasible inner validation. Its UniLasso implementation uses intercept-containing leave-one-out univariate predictions and a nonnegative second-stage regression with an intercept, checked against explicit leave-one-out refits. It is not replaced by a marginal-sign LASSO.

The additional producers are:

- Learned-group fits: `learned_pooling_validation.py`.
- Endpoint-grid fits: `tuning_boundary_validation.py`.
- Endpoint sensitivity: `tuning_boundary_sensitivity.py`.
- eQTL sensitivity: `eqtl_sensitivity.py`.
- eQTL stability: `eqtl_stability.py`.
- Computational checks: `computational_validation.py`.

The protocols, saved coefficients, per-seed or per-split summaries, and source snapshots accompany the revision.

In the complete-estimator studies, squared-error loss is normalized by the number of training observations. The revised group fits use the explicitly stated sparse-group objectives, fixed pilot information within a regularization path, and MOSEK through CVXPY; ordinary LASSO and the positive-penalty nonnegative UniLasso stage use coordinate descent. The UniLasso zero-penalty endpoint uses nonnegative least squares after centering its leave-one-out features and response, with the intercept restored afterward. For the added zero-penalty HCGL, FCGL, cooperative and ordinary Lasso cases, the objective reduces to separate responsewise least squares, retaining any gate constraints. We remove pinned-zero columns and use bounded-variable least squares for sign restrictions, or an ordinary least-squares solve when unrestricted. At mixing zero, the zero-solution scale is the dual norm of the corresponding row-HCGL or predictor-block FCGL group penalty. These endpoints introduce no ridge approximation and do not change the positive-mixing objective. Specifically, let $H = 2\boldsymbol{Y}^\top \boldsymbol{X}/T$ in the current training-standardized units. With the experiment's nonadaptive weights and $q_g = \sqrt{|\mathcal{C}_g|/G}$, the pure-group scales are

$$\lambda_{\max,H} = \max_g \max_{k\in\mathcal{C}_g} \frac{\|H_{k,:}\|_2}{q_g}, \qquad \lambda_{\max,F} = \max_g \max_j \frac{\|H_{\mathcal{C}_g,j}\|_2}{q_g}.$$

The unconstrained zero solution at these bounds remains feasible and optimal after adding the gate restrictions. The chosen solver must return an optimal solution with finite coefficients; the experiments do not silently change solvers after a failure. Preprocessing and optimization time are included in the benchmark timing summaries.

For dense inputs, forming marginal products costs $O(TNM)$; constructing the response correlation or Euclidean-distance information costs $O(TN^2)$ and uses $O(N^2)$ storage. Once the signs and weights are fixed, HCGL separates by response, whereas FCGL and cooperative norms couple coefficients inside their specified blocks. These observations describe structure, not an end-to-end

complexity guarantee for an interior-point solve. Actual runtime depends on dimensions, sparsity, tolerance and the number of fitted candidates.

S12.1 Measured path scaling

We run each seven-penalty path in a fresh process at four problem sizes, with three seeds per size and five conic estimators (60 completed paths). The design has blocks of four responses, within-block residual correlation 0.6, cut 0.7, $\alpha = 0.5$ and $\tau = 2$. Timing includes preprocessing, CVXPY canonicalization and path solves, but excludes imports and does not include cross-validation. Peak Windows working set includes the interpreter and numerical libraries. The host is an Intel Core Ultra 7 265 (20 logical processors, 31.46 GiB physical memory), running Windows 11; numerical libraries and MOSEK use one thread. Other research processes were active, so these are descriptive observed wall times rather than isolated hardware speed comparisons.
Table S24 gives the corresponding numerical results.

Table S24: Seven-point path scaling. Seconds are means (standard deviations) over three seeds in the clean-environment reproduction. Memory is the largest process peak over the three seeds, in MiB.

T	M	N	Method	Seconds	Peak MiB
80	12	12	HCGL, unconstrained	0.17 (0.03)	172
			HCGL, pooled/exclude	0.16 (0.07)	171
			HCGL, pooled/release	0.16 (0.02)	172
			FCGL, unconstrained	0.18 (0.02)	172
			Cooperative Lasso	0.25 (0.02)	177
120	60	24	HCGL, unconstrained	1.91 (0.15)	231
			HCGL, pooled/exclude	0.93 (0.09)	229
			HCGL, pooled/release	1.65 (0.04)	230
			FCGL, unconstrained	2.26 (0.16)	236
			Cooperative Lasso	6.51 (0.52)	295
160	120	48	HCGL, unconstrained	11.72 (0.47)	504
			HCGL, pooled/exclude	4.83 (0.41)	486
			HCGL, pooled/release	11.66 (0.16)	504
			FCGL, unconstrained	17.36 (0.87)	512
			Cooperative Lasso	52.29 (0.94)	990
160	240	64	HCGL, unconstrained	48.97 (2.49)	1223
			HCGL, pooled/exclude	14.61 (0.92)	976
			HCGL, pooled/release	46.53 (1.25)	1183
			FCGL, unconstrained	75.24 (5.77)	1307
			Cooperative Lasso	306.94 (36.16)	2453

The largest case has 160 observations, 240 predictors and 64 responses. These timings do not establish statistical recovery when dimension grows, nor do they cover nested tuning at these dimensions. The method, path and process boundaries are reported so that memory and timing cannot be mistaken for an end-to-end cost.

S12.2 Independent optimization and support checks

For three small-design seeds, five conic methods and penalty fractions $0.3, 0.03, 0.003$, we reconstruct 45 objectives independently using scalar row/block expressions and CLARABEL. The main MOSEK fits use primal, dual and relative-gap tolerances 10^{-8}. An initial reference request of 10^{-10} returns `optimal_inaccurate` in six cases; those statuses are retained. A uniform 10^{-9} reference request returns `optimal` for all 45, with maximum conic primal residual 8.16×10^{-10} and relative gap 8.99×10^{-10}. The largest relative difference in the directly evaluated objectives is 3.12×10^{-7},

and the largest coefficient difference is 3.05×10^{-4}. No reference failure is substituted for a main fit.

Objective agreement does not establish identical selected support. At coefficient tolerance 10^{-6}, the solvers disagree on selection in a mean 2.33% of cells across these 45 fits, with a maximum 10.42%; at 10^{-4} these are 0.278% and 2.083%. At 10^{-8} the corresponding proportions reach 16.08% and 65.97%. The zero/positive/negative classifications have the same disagreement rates. These denominators are all coefficient cells, not selected cells or true active cells. Consequently, small objective discrepancies cannot certify exact support or false-discovery rankings. The manuscript reports threshold sensitivity alongside prediction and unthresholded coefficient error; the audit is small-dimensional and does not certify every large-scale solution.

The archived simulation producer is `sign_pooling_simulation.py`. The robustness producer is `sign_pooling_robustness.py`. Figures are rendered by `exhibits.py`; their historical labels are corrected without changing the frozen input values. The corrected oracle diagnostic writes a separate `coop_benchmark_tolerance.csv`, and the exhibit producer selects it only through its explicit `--corrected-coop` argument. The older `eqtl_pipeline.py` and `eqtl_exhibits.py` retain descriptive reproduction, with an explicit warning on the superseded prediction path. The independent and date-score gate experiment is `weighted_gate_validation.py`. Research outputs and LaTeX builds are written outside the source checkout; reviewed manuscript assets are promoted separately. No unavailable financial replay is used to support the results in this revision.

Run integrity and provenance. The original learned-group study completed all 200 planned scenario–seed replications and eight methods. Its 8,800 coefficient arrays and all recorded validation losses were finite; an intentional preprocessing-leakage mutation was rejected. Four intermittent NumPy invalid-reduction warnings across the initial and mixing-extended runs prompted a separate audit. The installed CVXPY 1.9.2 sum atom reduces uninitialized floating-point memory during shape inference, a documented upstream defect.[1] Injecting a signaling NaN into this shape-only scratch memory reproduces the warning in the same model-construction path with finite model inputs. A scoped compatibility wrapper instead infers the shape using initialized integer zeros; it leaves numeric evaluation, canonicalization and solver settings unchanged. All 162 matched penalty solves produce bit-identical coefficients and intercepts after this repair, including under the injected-memory test. An additional complete replay of the 200 unmodified earlier mixing-extended replications reproduces their saved coefficients exactly. The four historical messages had no captured stacks, so they cannot be attributed individually with certainty retrospectively. The new endpoint-grid runs use the repair and treat RuntimeWarnings and NumPy invalid, division and overflow conditions as errors. Replay and diagnostic outputs never enter the statistical aggregates; no failed seed is discarded. Both endpoint studies completed: 1,080 primary fits across the six-design benchmark, 1,600 across the learned-group study, and 16,800 fixed-setting sensitivity fits. No new RuntimeWarning occurred. Independent reconstruction from saved coefficients reproduces every reported prediction, coefficient, support and gate metric to within 7.2×10^{-15}; all chosen settings match the minimum of their saved training-only CV surface.

The added pure-group and zero-penalty solvers were independently checked against 108 scalar-conic reference fits at CLARABEL tolerance 10^{-9}. All references were optimal; the largest relative objective discrepancy was 6.74×10^{-7}, with maximum coefficient difference 3.09×10^{-4} and zero primary constraint violations. Objective agreement still does not certify identical support at arbitrary numerical thresholds.

The run's `protocol.json`, `receipt.json`, source snapshots, per-seed CV surfaces and coefficient archives identify the fits. Paired sensitivity summaries are reconstructed from saved per-seed draws; `verify_revision_tables.py` and `verify_eqtl_tables.py` check the printed synthetic and eQTL tables against their numerical inputs. The endpoint extension, warning reproducer, independent objective checks and table reconstruction have separate source snapshots and audit receipts. A separate, stopped high-dimensional pilot is not included in these comparisons.

[1] CVXPY issue 3235: `https://github.com/cvxpy/cvxpy/issues/3235`.

S13 Additional gate validation

S13.1 Validation of the revised gate

The September gate validation contains 6,000 paired scenario–weighting evaluations: six cases, equal weighting and EWMA span 60, with 500 replications each. Each design has 240 dates, eight responses, one active predictor and one null predictor. Residual correlation is zero in the independent case, 0.8 in the overlapping case and one for exact copies. Other cases introduce unequal inception dates and predictor gaps, heteroskedastic noise, or a loading that changes from positive to negative. For that last case, the recovery target is the positive full-history average for equal weighting and the negative recent loading for span 60. Data and seeds are paired across gates and reused across cases and weighting modes. These are not 6,000 independent data panels. No fitted empirical coefficient is used as simulation truth.
Table S25 gives the corresponding numerical results.

Table S25: Revised gate validation at threshold one. Null admission means any admission on the null predictor; recovery means admission with the correct active sign. Percentages over 500 paired replications per row. Equal denotes equal weighting; 60 is the EWMA span.

		Independent gate		Date-score gate	
Design	Span	Null	Recovery	Null	Recovery
Independent residuals	Equal	33.4	100.0	30.6	100.0
Independent residuals	60	34.4	98.6	33.0	98.8
Correlated residuals	Equal	71.8	98.4	35.6	94.8
Correlated residuals	60	75.6	82.4	36.8	64.6
Duplicate responses	Equal	73.6	97.0	35.0	90.2
Duplicate responses	60	77.4	80.8	36.4	61.2
Unequal histories	Equal	65.6	96.2	31.6	91.2
Unequal histories	60	72.8	82.2	35.4	63.6
Heteroskedastic residuals	Equal	70.6	70.2	35.6	42.0
Heteroskedastic residuals	60	73.0	56.6	39.2	37.4
Changing sign	Equal	69.8	89.8	36.0	69.6
Changing sign	60	70.6	96.0	34.2	89.4

The dependence correction reduces null admission under overlap and exact duplication. At threshold one it remains a deliberately loose screen; a lower null-admission rate alone is not sufficient evidence of a better operating point. Table S26 therefore uses disjoint calibration and evaluation draws to compare approximately matched null-admission rates. The comparison retains heteroskedastic and changing-sign cases rather than selecting only favourable cells. Table S26 gives the corresponding numerical results.

Table S26: Separate threshold calibration and evaluation, EWMA span 60. The first 250 draws calibrate each threshold to 20% null admission; the other 250 assess null-admission and active-recovery percentages. This compares operating points, not a claim of exact error control.

	Independent gate		Date-score gate	
Design	Null	Recovery	Null	Recovery
Independent residuals	26.0	98.8	20.0	98.0
Correlated residuals	19.6	52.4	19.6	52.4
Duplicate responses	19.6	46.8	19.6	46.8
Unequal histories	20.8	53.6	19.6	52.0
Heteroskedastic residuals	19.2	22.8	19.2	25.2
Changing sign	20.0	77.2	19.6	77.2

Figure S1 compares the gate tradeoffs.

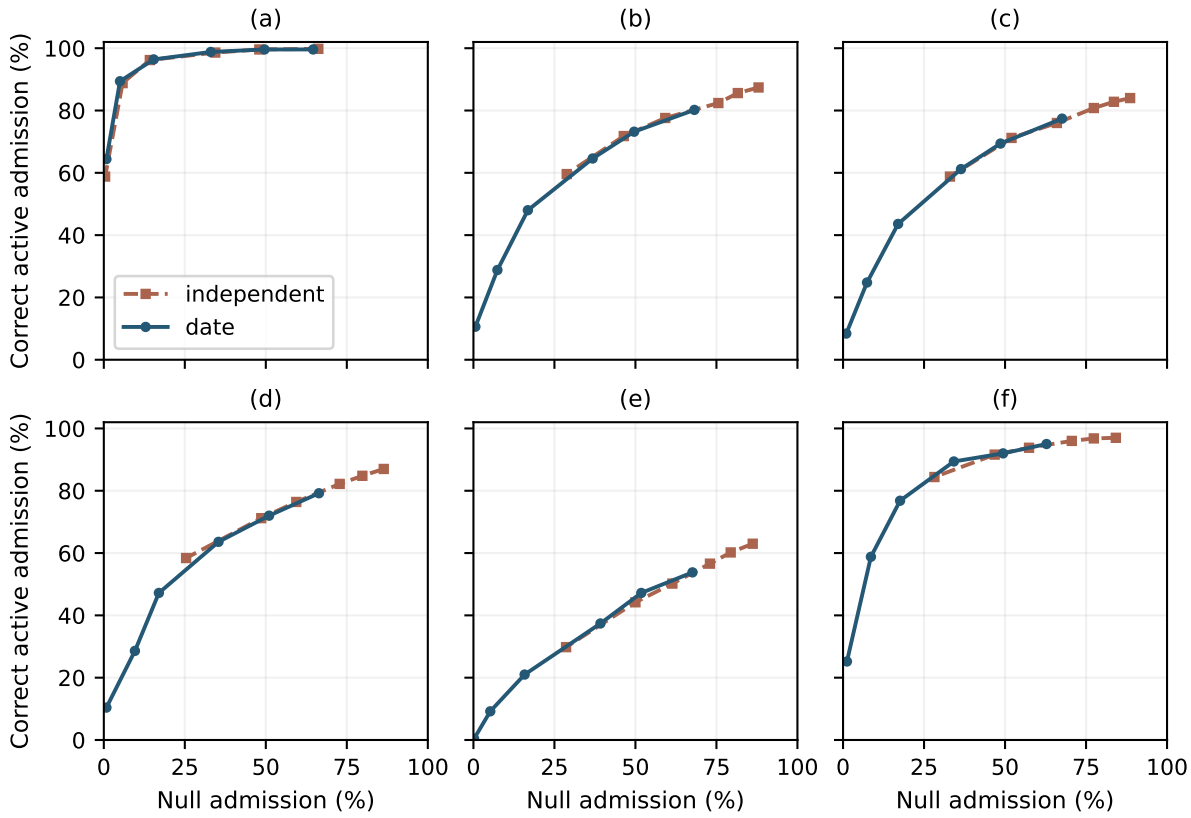


Fig. S1 Revised gate trade-offs at EWMA span 60. Dashed lines with squares show the independent-cell gate; solid lines with circles show the date-score gate. Panels: (a) independent residuals, (b) correlated residuals, (c) duplicate responses, (d) unequal histories, (e) heteroskedastic residuals and (f) changing sign. Each panel uses 500 paired replications and thresholds 0.5, 0.75, 1, 1.5, 2 and 3. Smaller null admission can trade off against active recovery; neither axis alone establishes superiority

At approximately matched null-admission levels, recovery is similar for the two screens in most cases. The evidence supports a correction for dependence and duplicated responses, not a general improvement in the full estimator. Threshold calibration in this experiment uses simulated null truth and is not a practical error-control procedure for real data. The finite-span and changing-sign cases are outside the asymptotic proposition.

Deterministic checks accompany the gate validation: repeating all members of a fixed cluster equally leaves the score unchanged; observation masks and original-grid recency weights are preserved. The complete-estimator comparison in Section 4.1 uses equal weights and independent dates, with its preprocessing repeated within each training fold.

S14 Optional target and adaptive-weight specification

When adaptive reweighting is off, $W_{kj} = A_k = A_{gj} = 1$. When it is on, the marginal magnitude weights are

$$W_{kj} \;=\; \frac{1}{\max\bigl(|\hat{\beta}_{j,\mathcal{C}(k)}|,\, \varepsilon\bigr)^{\gamma}}, \tag{S12}$$

with exponent $\gamma > 0$ and fixed stabilizer $\varepsilon > 0$. A zero-pinned cell is assigned placeholder weight one, which changes only a constant in the objective. Let $\mathcal{J}_k$ contain the predictors of row k that are not pinned, and $\mathcal{I}_{gj}$ the members of block (g, j) that are not pinned. The implemented group weights are

$$A_k = \left(\frac{\sum_{j\in\mathcal{J}_k} W_{kj}^2}{|\mathcal{J}_k|}\right)^{1/2}, \qquad A_{gj} = \left(\frac{\sum_{k\in\mathcal{I}_{gj}} W_{kj}^2}{|\mathcal{I}_{gj}|}\right)^{1/2}, \tag{S13}$$

with value one for an empty set. Thus adaptive group weighting can remain active at $\alpha = 0$; $\alpha = 1$ removes the group term.

This magnitude-based construction is motivated by adaptive penalties (Zou, 2006; Wang and Leng, 2008), but does not inherit their support or oracle results. Fixed ε bounds the weights. Moreover, a pooled marginal slope need not vanish on a zero partial coefficient, even with uncorrelated predictors when other members of its cluster are active. We make no support-recovery or oracle-distribution claim for these fitted estimators. The convex problem, numerical implementation and tuning protocol are described in Section S12.

S15 Optional masked and weighted gate

This appendix records optional software extensions of the gate. The complete fitted estimator evaluated in the main text uses complete, equally weighted observations; no missing-data fitting objective is proposed here. A validity indicator v_{tkj} is one only when response k and predictor j are both observed. Missing pre-inception returns must not be passed as observed zeros. For weights w_t, define

$$D_j = \sum_{t,k\in\mathcal{C}} w_t v_{tkj} x_{tj}^2, \qquad \widehat{b}_j = D_j^{-1} \sum_{t,k\in\mathcal{C}} w_t v_{tkj} x_{tj} y_{tk}. \tag{S14}$$

Equal weights remain the default of the optional `ewma_span=None` analytics. A finite span $s \geq 1$ uses $w_t = (1 - 2/(s+1))^{T-t}$ on the original row grid before masking. Gaps do not compress time. Inputs retain their caller's centering convention; the sign routine introduces no extra intercept.

For overlapping response histories, use the date score

$$u_{tj} = w_t x_{tj} \sum_{k\in\mathcal{C}} v_{tkj}(y_{tk} - \widehat{b}_j x_{tj}), \qquad \widehat{V}_j = \frac{n_{\mathrm{eff},j}}{n_{\mathrm{eff},j} - 1} \frac{\sum_t u_{tj}^2}{D_j^2}. \tag{S15}$$

Here $n_{\mathrm{eff},j} = (\sum_t w_t a_{tj})^2 / \sum_t w_t^2 a_{tj}$, where a_{tj} marks a date with the predictor and at least one response observed. This effective count is over dates, not response cells. A zero D_j supplies no slope

evidence and is excluded. For $D_j > 0$, the score-sandwich construction retains contemporaneous residual covariance (Cameron and Miller, 2015); using the effective-date multiplier is our finite-sample screening convention. We apply Equation (3) with $t_j = \widehat{b}_j/\sqrt{\widehat{V}_j}$. Fewer than two effective dates cannot pass a positive gate. This statistic is not asserted to have an exact Student distribution or multiplicity-adjusted error control. Independent dates are assumed; serial dependence would require an additional HAC or block treatment.

For a fixed cluster with masks and weights retained, repeating every member c times multiplies D_j and u_{tj} by c, so it leaves both the slope and Equation (S15) unchanged. In contrast, the archived independent-cell standard error treats copies as added information. Duplicating only one member of a heterogeneous cluster may change its weighting and hence the pooled estimand; no invariance is claimed in that case or for rerunning the clustering algorithm after duplication.

The design effect $1 + (m-1)\rho_e$ and effective member count $m/[1 + (m-1)\rho_e]$ describe the balanced equal-variance special case derived in Section 3, with ρ_e a marginal-regression residual correlation. Unequal histories and recency weights require the masked scores rather than a return-correlation adjustment. Missingness can also change the population being averaged; the score formula does not correct informative missingness. The consistency argument in Section 3 concerns complete observations and equal weights. They are not an extension to fixed-span EWMA: its effective temporal sample size stays bounded as more history is stored. Marginal-versus-conditional sign disagreement also remains possible after the variance repair.

These formulas describe screening only. Native-frequency routing, prior-target sign overrides and missing-data fitting require additional definitions and validation; they are outside the estimator studied here. The formulas also do not include additional influence terms from estimating column scales, and no calibrated variance assertion is made for that normalization.

S16 The closed-form SSR identity

S16.1 Pooled-slope identity

We derive the sign of each coefficient from a univariate slope, pooled across the responses that share a cluster. For predictor j, the pooled no-intercept slope across all N responses solves the first-order condition $\boldsymbol{x}_j^\top \sum_{k=1}^{N}(\boldsymbol{y}_k - \hat{\beta}_j \boldsymbol{x}_j) = 0$, which gives

$$\hat{\beta}_j \;=\; \frac{\boldsymbol{x}_j^\top \boldsymbol{y}_{\text{sum}}}{N\,\boldsymbol{x}_j^\top \boldsymbol{x}_j}, \qquad \boldsymbol{y}_{\text{sum}} \;\equiv\; \sum_{k=1}^{N} \boldsymbol{y}_k \;\in\; \mathbb{R}^T. \tag{S16}$$

Equation (S16) generalizes the single-response univariate slope of Chatterjee et al. (2025) to many responses, with the response sum $\boldsymbol{y}_{\text{sum}}$ in place of a single response. When the responses are grouped into clusters, we compute the slope against the cluster-sum response and broadcast it to every cluster member. The cluster-specific expression is Equation (2) in the main article. Equation (2) restricts Equation (S16) to the cluster index set, and recovers it when $\mathcal{C} = \{1, \ldots, N\}$. The cluster-pooled slope is the OLS slope of the cluster-mean response $\bar{\boldsymbol{y}}_{\mathcal{C}} = |\mathcal{C}|^{-1}\boldsymbol{y}_{\mathcal{C},\text{sum}}$ on $\boldsymbol{x}_j$.

Broadcasting a pooled slope gives every member the same permitted direction. This construction does not establish that their true partial coefficients have that direction. Positive response correlation alone is insufficient, and a cluster may contain both active and inactive responses for the same predictor.

S16.2 Residual identity

This appendix derives the residual sum-of-squares identity of Equation (S18) and gives the full t statistic.

Derivation. For predictor j, the pooled no-intercept slope $\hat{\beta}_j$ of Equation (S16) satisfies the first-order condition $\boldsymbol{x}_j^\top \boldsymbol{y}_{\text{sum}} = N\,\hat{\beta}_j\,(\boldsymbol{x}_j^\top \boldsymbol{x}_j)$. The pooled residual sum of squares aggregates the squared residuals of the N per-response fits at the common slope $\hat{\beta}_j$,

$$\begin{aligned}\text{SSR}_j &= \sum_{k=1}^{N} \big\|\boldsymbol{y}_k - \hat{\beta}_j\,\boldsymbol{x}_j\big\|_2^2 = \sum_{k=1}^{N} \Big(\|\boldsymbol{y}_k\|_2^2 - 2\,\hat{\beta}_j\,\boldsymbol{x}_j^\top \boldsymbol{y}_k + \hat{\beta}_j^{\,2}\,\boldsymbol{x}_j^\top \boldsymbol{x}_j\Big) \\ &= \|\boldsymbol{Y}\|_F^2 - 2\,\hat{\beta}_j\,\boldsymbol{x}_j^\top \boldsymbol{y}_{\text{sum}} + N\,\hat{\beta}_j^{\,2}\,\boldsymbol{x}_j^\top \boldsymbol{x}_j, \end{aligned} \tag{S17}$$

using $\sum_k \|\boldsymbol{y}_k\|_2^2 = \|\boldsymbol{Y}\|_F^2$ and $\sum_k \boldsymbol{x}_j^\top \boldsymbol{y}_k = \boldsymbol{x}_j^\top \boldsymbol{y}_{\text{sum}}$. Substituting the first-order condition $\boldsymbol{x}_j^\top \boldsymbol{y}_{\text{sum}} = N\,\hat{\beta}_j(\boldsymbol{x}_j^\top \boldsymbol{x}_j)$ into the middle term gives the closed-form identity

$$\text{SSR}_j = \|\boldsymbol{Y}\|_F^2 - N\,\hat{\beta}_j^{\,2}(\boldsymbol{x}_j^\top \boldsymbol{x}_j). \tag{S18}$$

This is an algebraic identity; it does not justify a variance formula under dependence or heterogeneous effects. The right-hand side requires only $\|\boldsymbol{Y}\|_F^2$ (computed once), the per-predictor inner product $\boldsymbol{x}_j^\top \boldsymbol{x}_j$, and the slope, so the full (N, M) statistic is one matrix product rather than a loop that forms the $T \times N$ residual matrix for each predictor.

Archived independent-cell statistic. The archived experiments use the following residual scale and screening statistic:

$$\hat{\sigma}_j^{\,2} = \frac{\text{SSR}_j}{N\,T - N}, \qquad \widehat{\text{SE}}\big(\hat{\beta}_j\big) = \sqrt{\frac{\hat{\sigma}_j^{\,2}}{N\,\boldsymbol{x}_j^\top \boldsymbol{x}_j}}, \qquad t_j = \frac{\hat{\beta}_j}{\widehat{\text{SE}}\big(\hat{\beta}_j\big)}. \tag{S19}$$

The cluster-pooled variant substitutes $|\mathcal{C}|$ for N and restricts the Frobenius norm to cluster columns. The denominator $NT - N$ is the historical screening convention retained to identify those results; no exact Student reference distribution is assigned to it. The SSR includes discrepancies between each response and the common pooled slope, including between-response slope heterogeneity. It does not estimate the cross-response covariance terms in Equation (12). In particular, independent full-model errors do not justify treating marginal residuals as independent cells.

The revised statistic uses Equation (4), which squares the sum of residual scores within a date. For a prespecified cluster, complete observations, equal weights and fixed analysis scales, it is the heteroskedasticity-robust slope variance for regression on the cluster-mean response, with the stated finite-sample multiplier. With sample-estimated column scales, the following proposition uses its positive score-scale limit without identifying it as the full sampling variance of the normalized slope. It is this statistic, rather than Equation (S19), that the proposition covers.

S17 Clustering conventions

The partition can be supplied externally or estimated from training responses. The revised fitted-estimator experiments use $D_{kl} = \sqrt{2(1 - C_{kl})}$ with $\boldsymbol{C} = \text{corr}(\boldsymbol{Y})$, the Euclidean distance between centered response vectors normalized to unit length, and Ward linkage on these distances. This interpretation requires correlations computed from a common observation grid. Arbitrary pairwise-missing correlations need not define a Euclidean configuration, and a monotone transformation of the distances does not in general preserve Ward's tree. The cut is a predeclared fraction of the final Ward merge height, 0.5 in the synthetic benchmark and 0.7 in the nested eQTL analysis. Under this rule a fraction of one produces a single cluster, while small positive cuts need not produce all singletons when distances are zero. A target number of clusters is an alternative cut rule and is stated when used. The cut fraction is a user-selected hyperparameter, so the partition depends

on both data and tuning choices, and none of these constructions guarantees agreement of partial coefficient signs within a discovered cluster.

The companion software manuscript and the synthetic teaching examples of Section S10 use the package's own convention, Ward linkage applied to $1 - C_{kl}$ with a distance cut at a fraction of the largest pairwise distance. That transform is a squared-distance heuristic rather than a Ward sum of squared deviations, and its partitions are not interchangeable with those of the revised protocol. The record of the historical experiments and of the superseded eQTL analysis is kept with the replication materials in Online Resource 2.

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